Journal of Banking and Finance 162 (2024) 107126

Contents lists available at ScienceDirect

Journal of Banking and Finance

journal homepage: www.elsevier.com/locate/jbf

The green corporate bond issuance premium ☆

John Caramichael [a], Andreas C. Rapp [b,*]

[a] The Demex Group, 1100 15th Street NW, Suite 400, Washington, DC 20005, United States of America
[b] Federal Reserve Board, 20th Street and Constitution Ave NW, Washington, DC 20551, United States of America

ARTICLE INFO

JEL classification:
C33
G15
G18
G23
G28
Q54
Q56

Keywords:
Green bonds
Corporate bonds
Bond issuance
Green bond premium
Green finance
Sustainable finance
Climate finance

ABSTRACT

We study a global panel of green and conventional corporate bonds to assess the borrowing cost advantage at issuance for green bond issuers. We find that, on average, green corporate bonds have a yield spread that is between 3 and 8 basis points lower relative to conventional bonds, depending on the regression specification. We link this borrowing cost advantage, or "greenium," to demand pressure at issuance, highlighting a key mechanism through which the greenium is allocated. We find that a significant greenium emerges only as of 2019, coinciding with the growth of the sustainable asset management industry following EU regulation. While green bond governance and external review appear to matter for the greenium, the credibility of the underlying projects has little impact. Instead, the greenium is unevenly distributed to large, investment-grade issuers, primarily within the banking sector and developed economies. These findings have implications for the role of green bonds in incentivizing meaningful green investments throughout the global economy.

1. Introduction

Green corporate bonds have grown in importance in recent years and now account for nearly 6% of global corporate bonds outstanding, up from less than 1% in 2014 (Fig. 1). Green bonds are similar to conventional corporate bonds but contain provisions that direct the funding raised from their issuance toward climate- and environment-friendly (green) projects. Although there are several potential benefits of issuing green bonds – such as catering to investors' changing attitudes toward sustainable investing, retaining environmentally conscious employees, or creating a long-term platform for stakeholder engagement – it is unclear whether and how green bonds benefit issuers on the margin.

In this paper, we focus on a firm's potential direct benefit of issuing green bonds by testing whether these bonds have lower yields at issuance. We focus on the yield at issuance because the primary market determines the actual interest rate paid by issuers to borrow funds. A lower borrowing cost relative to conventional bonds, or "greenium," could offer a direct incentive to firms that seek to invest in green

☆ We thank the Editor, Geert Bekaert, an anonymous Associate Editor, and an anonymous referee, whose insightful comments have improved our paper. For their continued interest and guidance at various stages of the project, we are indebted to Ricardo Correa and Frank Warnock. For valuable suggestions, we thank Michele Dathan, Anil Jain, Yuriy Kitsul, Jac Kragt, Gordon Liao, Andrew McCallum, Viktors Stebunovs, Emre Yoldas, Tony Zhang, Alex Zhou, and participants at the Division of International Finance Policy Workshop, the Georgetown-IDB Green Bond Methods Seminar, the World Bank DMF Sustainable Bond Webinar, and the GBTP Research Seminar Sessions. When this paper was initially written, Caramichael was a senior research assistant at the Federal Reserve Board. We have nothing to declare. The views expressed in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or any other person associated with the Federal Reserve System.

* Corresponding author.
E-mail addresses: john.caramichael@thedemexgroup.com (J. Caramichael), andreas.c.rapp@frb.gov (A.C. Rapp).

https://doi.org/10.1016/j.jbankfin.2024.107126
Received 10 January 2023; Accepted 1 March 2024
Available online 13 March 2024
0378-4266/Published by Elsevier B.V.

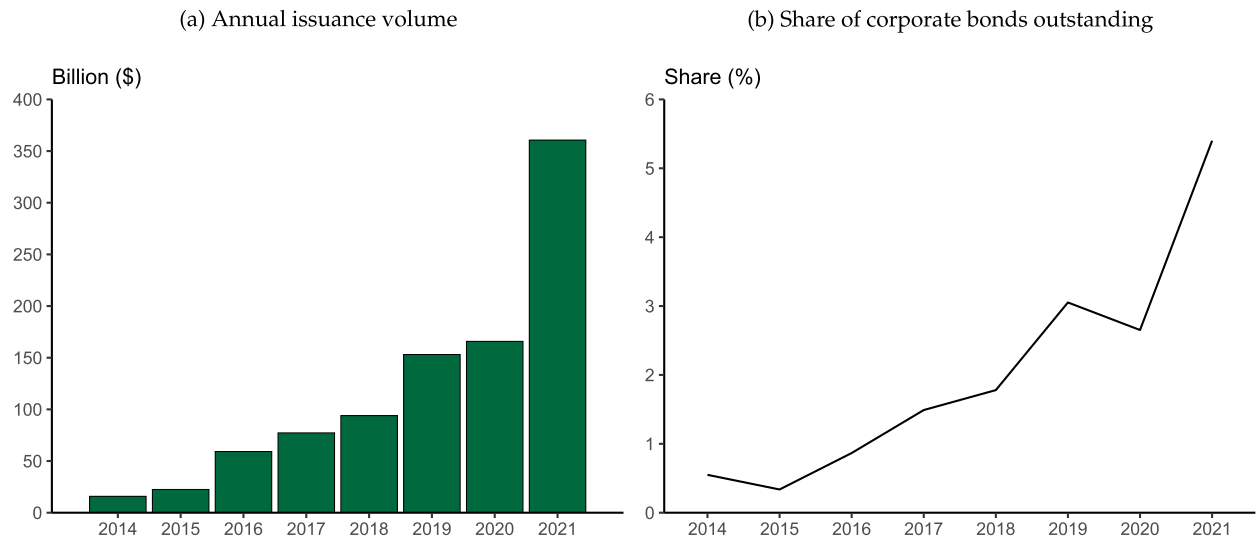

Fig. 1. Growth in green corporate bonds. Panel (a) is the annual issuance volume of the global green corporate bond market. Panel (b) is the notional share of green corporate bonds relative to the notional size of the total corporate bond market at year-end.
Source: Bloomberg Back Office Fixed Income Corporates Reference Data (2021).

projects.[1] However, such an incentive critically depends on whether the greenium is economically meaningful, on the potential drivers behind the greenium, and on how the greenium is distributed across different countries, industries, firms, and projects.

We therefore seek to answer the following questions: First, is there empirical evidence for a significant, average, corporate greenium at issuance? If so, does it carry sufficient economic significance to incentivize green investment? Second, does the corporate greenium vary over time? How does the potential time variation in the greenium connect with the maturation of the green bond market? And can the greenium be linked to increased demand for green bonds, possibly from the growth of the sustainable asset management industry following EU regulatory changes? Third, how is the greenium distributed across bonds and issuers? In particular, is the greenium distributed evenly across firms, or is it tilted to certain countries, industries, or issuers?

To address these questions, we leverage a comprehensive global panel dataset with information from international primary bond markets. Our sample contains 1,169 green and 129,043 conventional corporate bonds issued between 2014 and 2021. These bonds, denominated in 23 currencies, were issued by 12,736 firms in both advanced and emerging market economies.

Based on this comprehensive dataset, we primarily use a fixed-effects regression approach comparable to that of Baker et al. (2022) across an unbalanced panel of corporate bonds to understand the drivers of variation in bonds' yield spread at issuance.[2] However, as noted by Larcker and Watts (2020), fixed-effects models can produce spurious inferences if the regression controls are not adequately specified. To alleviate this concern, we corroborate our key results through matched sample regressions, extending the approach of Flammer (2021).

Our analysis yields five central findings: First, on average, U.S. dollar- and euro-denominated green corporate bonds offer a small direct borrowing cost advantage of between 3 basis points and 8 basis points over our sample period, depending on the regression specification. Relative to the average yield spread in the corresponding regressions, this reflects a 2% to 7% reduction in the borrowing cost to the firm. This borrowing cost advantage is economically meaningful, but it is likely small or potentially negative when taken net of fees and compliance costs (certification, monitoring, and reporting), particularly for complex green projects and small or first-time issuers. Our findings are robust to the inclusion of nonlinearities and time-varying issuer fixed-effects, as well as to the use of subsamples and matched samples that better balance the control group of conventional bonds with our green bond sample. They are also distinct from a green bond issuer effect, or "green halo," which is a hypothesized benefit of issuing a green bond that lowers an issuer's overall borrowing costs across all bonds, both green and conventional.

Second, we find that a statistically significant greenium emerges only as of 2019 at about 15 basis points and then tightens to about 10 basis points and 8 basis points in 2020 and 2021, respectively. During this time, corporate bond investors increased their demand for this type of bond, and European Union (EU) officials introduced broad sustainable finance policies. While EU regulation appears to have supported the development and strength of the green bond market in Europe, our results do not establish a causal link between the regulation and the emergence of the greenium in 2019.

Third, we find that the greenium is linked to two proxies of demand pressure, oversubscription and green bond index inclusion, highlighting mechanisms through which the greenium can be allocated as demand for green bonds outpaces supply. Our findings reveal that green bonds are more oversubscribed than comparable conventional bonds and that strong investor demand largely explains green bonds' borrowing cost advantage. Our results point to a more pronounced greenium through demand pressures in euro-denominated green bonds as well as U.S. dollar green bonds of foreign issuers. As the emergence of a significant greenium coincides with the rapid growth of the sustainable asset management industry, excess demand for green bonds at issuance appears to be a potentially temporary driver of the green bond borrowing cost advantage.

Fourth, while U.S. dollar- and euro-denominated green bonds capture comparable greeniums, the greenium is, on average, allocated primarily to local euro and foreign U.S. dollar issuers. In our sample, there appears to be no significant average greenium in the domestic dollar market and in the non-euro area market more generally, perhaps be-

[1] The green bond yield premium, or "greenium," is defined as the difference in issuing yields between a green bond and a comparable conventional bond, holding other factors constant. Negative values reflect a borrowing cost advantage of the green bond over comparable conventional bonds.

[2] We choose to primarily use a fixed-effects regression approach because a matching approach drastically reduces the number of green bonds that can be investigated and skews the sample to issuers with strong capital market access and the ability to issue comparable corporate bonds frequently. This leads to an underrepresentation of small and emerging market issuers.

cause of the more stringent transparency requirements that European issuers and financial service providers are subjected to.

Fifth, while green bond governance and external review appear to matter for the greenium, the credibility of the underlying projects does not have a significant impact, suggesting that the greenium is primarily allocated to green bonds that are sufficiently green beyond some minimum threshold. Instead, the greenium is unevenly distributed to large, investment-grade issuers, primarily within the banking sector and developed economies, which points to firms that can meet typical index inclusion criteria.

Our paper adds new insights to the green bond literature, particularly related to studies on the greenium. Relative to the existing literature, our contribution strictly focuses on the greenium from the firm's perspective. Existing empirical evidence on the greenium's size and direction is mixed and best categorized in studies that focus on either the primary or secondary bond market.[3] For the latter, see, for example, Ehlers and Packer (2017) and Gianfrate and Peri (2019) for the greenium at issuance in the early years of a growing market; Partridge and Medda (2020), Larcker and Watts (2020), and Baker et al. (2022) for papers on the greenium in the U.S. municipal primary green bond market; Deng et al. (2020), Wang et al. (2020), Hu et al. (2022), Li et al. (2022), and Tang et al. (2023) for the greenium in the Chinese primary green bond market; Kapraun et al. (2021) for the greenium in an international sample across issuer types and market segments; and Wang and Wu (2023) for an analysis of green primary market book-building data. While the greenium in the secondary market is paid by investors to bondholders and is often estimated at only a few basis points, the greenium at issuance determines the borrowing cost advantage captured by the issuer, and its estimates typically fall into a wider range. However, it is difficult to parse and compare the existing results across papers because they differ widely in their sample periods, green bond market segments, geographical coverage, and empirical methodologies.

Our paper also contributes to this literature by collecting one of the most comprehensive global panels to date covering primary corporate bond markets from advanced and emerging market economies. We extend the literature by employing a careful empirical approach that allows us to tackle at least two new angles. For one, we capture time variation in the primary market greenium, documenting that a significant average greenium only emerged as of 2019. For another, we link the greenium to two proxies of excess demand, characterizing the borrowing cost advantage as the result of a mechanism through which firms respond to the high demand for their green bonds. This channel appears particularly relevant given recent empirical evidence that highlights the growing competition for primary market allocations from institutional investors in order to avoid the substantial transaction costs associated with buying the same position in the secondary bond market Nikolova et al. (2020); Flanagan et al. (2021). Moreover, using textual data from green bonds' use of proceeds, we also construct a new measure of green credibility that captures whether the bond has been used to refinance a firm's existing projects, adding to Kapraun et al. (2021), who focus on the investor's perspective on green credibility and the greenium. More broadly, our findings also add to the strand of the literature that links better corporate social responsibility performance to a reduction in firms' debt funding costs (Ge and Liu, 2015; Oikonomou et al., 2014; Bauer and Hann, 2010; Stellner et al., 2015).

Our findings also contribute to the broader literature on the issuance of securities in bond markets. We highlight that demand pressure at issuance is a key mechanism through which the greenium appears to be allocated. When firms experience high demand for their debt, they can raise the price of the debt (lower the yield), supply more debt, or both (see, for example, Ma (2019) and Mota (2020) for bonds, and Bolton et al. (2013) and Baker and Wurgler (2002) for general external financing). An elastic supply in response to high demand makes it difficult to econometrically identify excess demand for a firm's debt simply by regressing the quantity supplied on yield spreads. Siani (2022) overcomes this issue for conventional bonds by using variation in bond supply on the issuance day to show that firms respond to lower yields by increasing the quantity supplied of bonds. In our setting, the supply of green bonds for a given firm is rather inelastic to demand, because the notional amount of the green bond is limited by the opportunities for green investment available to the issuer. Increasing the size of the green bond beyond these opportunities would call into question the "greenness" of the bond. Instead, it is more feasible for the issuer to capture the high demand with a lower bond yield. This setting makes oversubscription ratios in green bonds a good proxy for excess demand and allows us to capture how issuers can benefit from demand pressure at issuance. We further confirm this benefit using bond index inclusion as another proxy for strong demand, contributing to the literature that shows that increased demand from mutual funds and exchange-traded funds (ETFs) benchmarked to a given bond index can impact bond pricing (Ye, 2019; Dannhauser, 2017; Holden and Nam, 2019), issuer behavior (Calomiris et al., 2022; Dathan and Davydenko, 2020), and exchange rates (Raddatz et al., 2017; Pandolfi and Williams, 2019). More broadly, our findings also add to the literature that studies an on-the-run liquidity premium typically observed in U.S. Treasury bonds (see, for example, Krishnamurthy (2002) and Vayanos and Weill (2008)), as the greenium at issuance documented in our paper is distinct from concerns about (expected) liquidity but still goes in the same direction as the on-the-run liquidity premium.

Lastly, our paper speaks to the literature on foreign currency bond issuances and corporate debt in emerging markets (see, for example, Bruno and Shin (2020), Beltran et al. (2017), Alfaro et al. (2019), Avdjiev et al. (2014), Ayala et al. (2017)). In a recent paper, Calomiris et al. (2022) show that emerging market corporate issuers opt for large-denomination foreign bond issuances that are eligible for inclusion in international bond indexes in an attempt to cater to institutional investors. Our findings connect with this observation, pointing to a greenium for firms whose bonds can meet typical index inclusion criteria – namely, issuers with an ability to regularly issue large, investment-grade rated bonds, primarily within the banking sector. This has implications for the role of green bonds in incentivizing meaningful green investments throughout the global economy, particularly in emerging market economies.

The remainder of the paper proceeds as follows: In Section 2, we discuss the growth of the green bond market, describe our dataset and yield spread computation, and outline potential motivations for issuing green bonds. In Section 3, we describe our independent variable construction and regression methodology. In Section 4, we present and analyze our empirical results. In Section 5, we discuss the broader significance of our findings and conclude the paper.

2. Background and bond sample

Before we begin our analysis, we first review key concepts and developments within the green bond market at large. We further detail the sample construction and discuss firms' motivation for issuing green bonds.

2.1. The green bond market

Green bonds are one of the most recognized instruments for the financing of green projects. While the stock of green loans is also growing, most green debt is in the form of green bonds. Like conventional bonds, green bonds are long-term fixed-income debt instruments. They generally have the same seniority, recourse, and rating as an issuer's

[3] For research on the greenium in secondary bond markets, see, for example: Hachenberg and Schiereck (2018); Immel et al. (2021); Nanayakkara and Colombage (2019); Karpf and Mandel (2018); Bachelet et al. (2019); Hyun et al. (2021); Lau et al. (2022); Pietsch and Salakhova (2022); and Zerbib (2019).

conventional bonds. The main difference between green and conventional bonds is that the proceeds of a green bond are earmarked for climate- and environment-friendly projects.[4] This typically includes investments in clean energy, energy efficiency, green buildings, or electrified transportation.[5] Green bonds rely on third-party certification or the trust of investors to assure that the bonds' proceeds are channeled toward the intended green investments. In short, green bonds are conventional bonds packaged with a "green promise."

Green bonds currently lack a universal global regulatory framework. Instead, they are customarily structured to align with the Green Bond Principles (GBP) published by the International Capital Markets Association.[6] The principles provide voluntary best-practice guidelines for the selection, management, evaluation, and disclosure of green projects when issuing a green bond. The principles recommend (but do not require) third-party certification for green bonds. As alignment with the principles is voluntary, the proceeds of some green bonds may not be invested in meaningful green projects, a practice that reflects broader trends of "greenwashing," or misusing the green label, in the sustainable finance industry.

The overall green bond market, including sovereign, quasi-sovereign, municipal, and corporate issuers, has grown rapidly in recent years, with issuance reaching $556 billion in 2021. This is a substantial increase from only $31 billion in 2014, reflecting a compound annual growth rate of about 50% for the entire market. While green bonds have been issued in many countries, European issuers have been the most active, followed by U.S. issuers, Chinese issuers, and supranational issuers. About three-quarters of annual total green bond issuance is denominated in the euro or the U.S. dollar.

Between 2018 and 2019, the green bond market saw a significant surge in issuance. During this time, regulators began to introduce broad sustainable finance policies, particularly in Europe, and there was significant growth in assets under management (AUM) by green bond funds, as charted in Fig. 2. The growth in AUM is driven almost exclusively by green bond funds domiciled in Europe, which account for more than 90% of green bond fund AUM.

During this period of persistent growth, policymakers, market participants, and financial journalists acknowledged the potential for green bonds to fund green investments and have a discernible positive impact. However, they also drew attention to some of the shortcomings of the current market. Key issues include greenwashing, the lack of a universal governance and certification framework, and the significant compliance costs associated with certification, issuance, and reporting.[7] Critics question whether the green bond market actually influences issuers' investments or if green bonds instead fund green projects that would have otherwise received funding through conventional channels.

2.2. Sample construction

Our focus is on green corporate bonds, which now account for about two-thirds of global issuance. Private-sector green bond issuers tend to be large, mature firms or firms with strong access to debt capital markets. This is especially true of firms issuing green bonds in a foreign currency to tap international green capital markets. Issuers also tend to be from relatively less carbon-intensive industries and greener than their industry peers (Flammer, 2021).

We obtain our global corporate bond sample by collecting bond-level data from Bloomberg Back Office (2021), our primary data source. For the period from 2014 through 2021, we compile a global panel of 129,043 conventional corporate bonds and 1,169 green corporate bonds. These bonds were issued by 12,736 corporations – of which 387 are green bond issuers – from advanced and emerging market economies. For the bonds in our sample, Bloomberg Back Office (2021) data provide detailed information on the characteristics of each bond, such as the issue price and size, issue and maturity date, currency and seniority, and potential option features. We obtain bonds' credit rating from Bloomberg BVAL (2021) and the variable to identify green bonds from BloombergNEF (2021). The Bloomberg green bond identifier strictly requires referenced bonds to be aligned with the first principle of the GBP, which states that the green bond's use of proceeds for environmental projects should be "*appropriately described in the legal documentation of the security*." We further supplement the Bloomberg issuance data with additional bond data from Refinitiv Workspace (2021) and Refinitiv Eikon (2021).

To construct our sample, we begin by selecting all conventional and green corporate bonds and medium-term notes from both green and conventional issuers.[8] We then require a bond to be a fixed- or zero-coupon bond with a notional amount of at least $500,000 at issuance and between 1 and 30 years until maturity.[9] Furthermore, we only consider bonds issued by private-sector corporations but allow state-owned enterprises (SOEs) in our sample.[10] We do not consider bonds directly issued by sovereigns, supranational entities, or municipalities.

We require green bonds to be issued in either the euro or the U.S. dollar because these green bond markets are the deepest and most liquid. While this restriction reduces the number of green corporate bonds in our sample by roughly half, the remaining bonds account for 70% of the green corporate bond issuance volume over our sample period.[11] We also drop green bonds if their ultimate parent does not have at least one conventional bond in our sample, which reduces the number of

[4] Conventional bonds can still be used to finance green projects. Green bonds are distinguished by the specification of the use of proceeds in the bond prospectus.

[5] While this paper focuses on the reduction of greenhouse gas emissions to mitigate climate change, green bonds may pledge investment to achieve a variety of aims not limited to reducing emissions, such as increasing water efficiency or supporting biodiversity.

[6] See International Capital Markets Association (2021).

[7] For further discussion on the shortcomings of green bonds, see Ritchie et al. (2021) and Stubbington and Nauman (2020). For discussion on greenwashing in the sustainable finance industry, see Quinson (2021a,b), The Economist (2021), Fancy (2021), and Temple-West and Palma (2022). For a critical assessment of the changes in carbon emissions from green bond issuers, see Ehlers et al. (2020).

[8] We use bonds with FCI codes starting with DB or DT. We do not consider bonds with attached warrants, convertible bonds, asset- or mortgage-backed securities, or structured notes and products.

[9] For some emerging market currencies, the availability of sovereign yield curve data (required to construct yield spreads) dictates slightly shorter maturity ranges.

[10] Many emerging market issuers with consistent access to international bond markets have some form of government backing; this includes SOEs. We retain SOEs because excluding them would reduce our global sample by roughly 20%. Keeping them does not qualitatively affect our results. We add regression controls on the issuer- and issuer-times-country levels to account for potential differences in SOE bonds.

[11] The Chinese yuan ranked third in issuance volume over our sample period. However, we exclude yuan-denominated green corporate bonds due to the following reasons: First, only 15% of green corporate bond issuance was denominated in yuan (16% in all other currencies combined), while euro- and U.S. dollar-denominated issuance stood at 42% and 27%, respectively, reflecting differences in market depth over our sample period. Second, U.S. and European bond markets are arguably more integrated with each other than they are with the Chinese market (Miranda-Agrippino et al., 2020), making a more homogenous comparison of the greenium at issuance more likely. Third, China's domestic green bond standards were only harmonized in definitions and practices with international standards over time, reaching close to full harmonization only after our sample period in mid-2022. In combination with strong incentives for green funding markets from the Chinese government, the ability to choose between local and international standards may affect the composition of issuers and investors at issuance (see, for example, Lin (2023)), an aspect not easily addressed with our data.

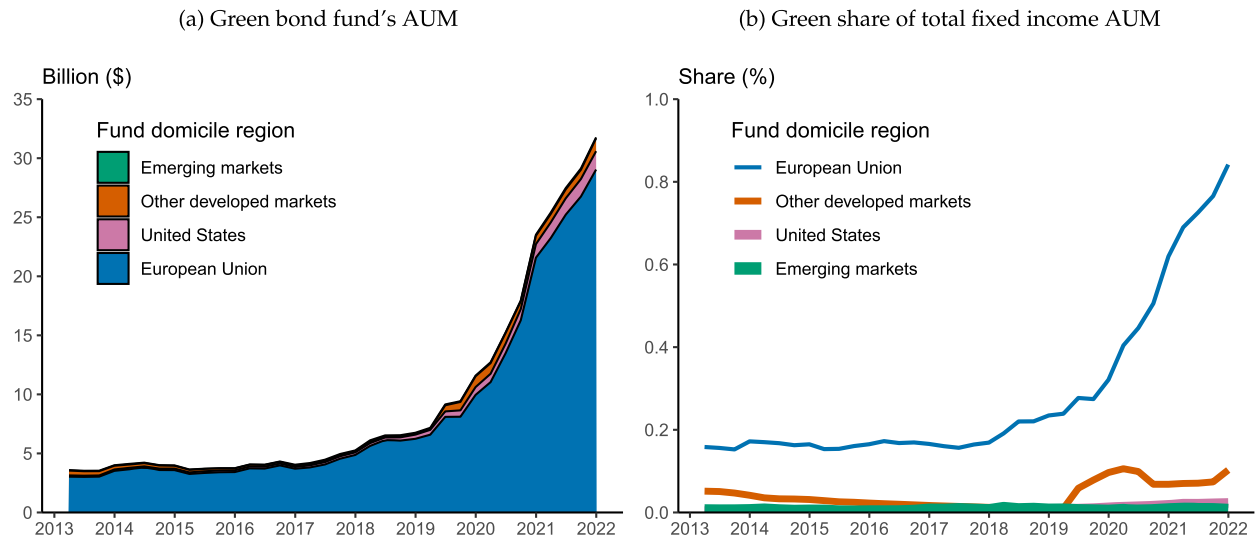

Fig. 2. Green bond funds' assets under management. Panel (a) is the assets under management (AUM) of green bond funds by fund domicile region. Panel (b) is the share of green bond fund AUM relative to each domicile region's total fixed income fund AUM. The data cover the period from 2014 through 2021 at a quarterly frequency.
Source: Morningstar, Inc. (2021).

green bonds by 5%.[12] Lastly, for conventional bonds, we include only bonds denominated in currencies where there is at least $10 billion in total issuance over the sample period to ensure that the conventional bond market has sufficient issuance volume for a meaningful comparison. The end result is that we include conventional bonds from 23 currencies. A breakdown of our sample by currency is provided in Table A1 in the Appendix. After we screen all remaining bonds for potential data issues concerning the price and the yield spread (see Subsection 2.3) we end up with a final sample of 130,212 green and conventional corporate bonds.

2.3. *Motivation for issuing green bonds: the greenium*

Firms that issue a green bond may receive several direct and indirect benefits that potentially incentivize green investment.[13] Issuing a green bond may effectively lower the interest rate paid on the bond relative to conventional bonds. If a firm chooses to issue a green bond, it may attract new investors interested in sustainable investment, thereby increasing demand for the bond. Should the added demand push the green bond's yield lower than that of a comparable or hypothetical conventional bond, this yield difference represents the green premium, or greenium.[14,15]

To construct our dependent variable, we first calculate the exact yield to maturity on issuance day using each bond's issue price, coupon rate, coupon frequency, and day-count convention. We focus on the primary market yield (that is, the yield at issuance) instead of the secondary market yield because the primary market determines the actual interest rate paid by issuers to borrow funds.[16] Specifically, the demand for bonds supplied in the primary market comes from institutional investors who are more likely to allocate money to one of several comparable primary market bond offerings rather than choose between a given issuer's new primary market issue and its outstanding bonds trading in the secondary market.[17]

To account for interest rate differences across currency regions, we then proceed to calculate each bond's yield spread at issuance by taking the difference between the yield to maturity and the linearly interpolated maturity-matched government bond yield curve for the given bond's currency on its date of issuance. Lastly, we take several steps to eliminate outliers and data errors.[18] By focusing on yield spreads, we are able to control for the differences in interest rate environments across currencies.[19] Summary statistics for the bonds in our sample are provided in Table 1.

3. Variable construction and empirical approach

The independent variables in our dataset are based on traditional financial securities data, such as the price, credit rating, seniority, and maturity of each bond, as well as metrics more specific to green bonds, such as whether a green bond experienced strong investor demand at issuance or whether it adheres to voluntary green bond standards. This section details the variable construction and the paper's regression approach.

[12] Data on issuers' ownership structure are from Bloomberg Global Corporate Structure (CAST) (2021).

[13] For a summary of common explanations for why investors would accept a lower yield on a green bond relative to a comparable conventional bond, see Section A1 in the Appendix.

[14] Alternatively, an issuer may respond to the increased demand by issuing more debt at the same interest rate. In practice, this is constrained by the limited number of eligible green projects available to the issuer.

[15] We do not have information on the costs associated with certifying, issuing, monitoring, and reporting over the bond's lifetime. But these costs can be high, especially for complex green projects and small or first-time issuers. Thus, it may take several green bond issues or a very large greenium for the greenium's borrowing cost advantage to break even.

[16] We also avoid the comparison of a bond's yield at issuance with secondary-market yields because the latter may not accurately reflect borrowing costs for corporate issuers. This is because liquidity conditions in secondary markets vary significantly over time (Bao et al., 2011) and across international markets (see, for example, Houweling et al. (2005); Hameed et al. (2019)).

[17] Institutional investors face substantial transaction costs when acquiring large bond positions in the secondary market, explaining their strong preference for purchasing bonds in the primary market (see, for example, Nikolova et al. (2020); Flanagan et al. (2021)).

[18] Specifically, we drop bonds with yield spreads greater than 10%, as these bonds are generally considered distressed. We also drop bonds with prices at issuance of less than $90 and greater than $250, as these are likely distressed, data errors, or outliers.

[19] That said, we do not have information on firms' currency hedging behavior.

Table 1
Bond sample summary statistics.

	Observations	Mean	S.D.	25th	Median	75th
Green bonds						
Yield spread (bp)	1,169	149.83	128.27	75.31	100.31	190.10
Amount issued ($M)	1,169	419.39	488.35	29.04	370.23	598.20
Coupon (%)	1,166	2.01	1.90	0.45	1.50	3.05
Years to maturity	1,169	7.76	4.08	5.00	7.00	10.00
Rating	735	7.55	3.25	5.00	8.00	9.00
Level	1,169	-89.00	150.15	-162.39	-113.06	-8.08
Slope	1,169	-3.82	44.49	-19.69	-6.39	19.67
Curvature	1,169	-0.10	9.16	-4.17	-0.72	3.59
Realized vol. (%)	1,169	5.89	3.15	3.77	5.23	7.11
IG index spread	1,169	118.63	29.50	96.00	111.00	132.00
HY index spread	1,169	450.95	110.12	373.00	405.00	517.00
Oversubscription	481	3.81	2.14	2.25	3.33	4.62
Conventional bonds						
Yield spread (bp)	129,043	137.27	168.31	43.01	97.80	198.39
Amount issued ($M)	129,043	256.30	623.93	16.94	75.96	300.00
Coupon (%)	126,443	3.59	2.74	1.50	3.00	5.00
Years to maturity	129,043	6.14	4.68	3.00	5.00	8.00
Rating	46,201	7.65	3.90	5.00	7.00	10.00
Level	129,043	-23.17	192.73	-120.11	-23.40	69.49
Slope	129,043	2.76	52.13	-21.11	-2.14	24.66
Curvature	129,043	-0.22	10.68	-5.95	-0.69	4.99
Realized vol. (%)	129,043	5.99	3.79	3.57	5.25	7.44
IG index spread	129,043	125.42	29.40	107.00	121.00	134.00
HY index spread	129,043	464.60	118.54	382.00	419.00	519.00

Summary statistics for the full bond sample, separating green and conventional bonds. Bond types are compared across the following metrics: *Yield spread (bp)* is bonds' yield spread at issuance, reported in basis points. *Amount issued ($M)* is bonds' issue size and reported in U.S. dollar millions. *Coupon (%)* is bonds' coupon (for fixed coupon bonds only) and reported in %. *Years to maturity* represents bonds' time-to-maturity on the issuance day, reported in years. *Rating* represents a bond's credit rating (for rated bonds only), ranging from AAA = 1 to C = 21. We define a bond's credit rating as the mean of credit ratings, where available, from Moody's, Standard & Poor's, and Fitch, and then rounding down. *Level, slope,* and *curvature* of the sovereign yield curve for a given bond's currency region on issuance day, calculated between 1 and 10 years to maturity using a principal component analysis. *Realized vol. (%)* is the 30-day realized volatility of the 10-year sovereign bond yield for a given bond's currency region on issuance day. *IG index spread* and *HY index spread*, respectively, represent the option-adjusted bond index spread of the ICE BofA global investment grade corporate bond index (G0BC) and the ICE BofA global high-yield corporate bond index (HW00) on a given bond's issuance day. Available only for a subset of green bonds in the full sample analysis, we also report *Oversubscription*, which represents green bonds' oversubscription ratio, defined as the ratio of the notional amount of bond orders to the actual amount of debt that is issued. The bond sample and variable construction are detailed in Sections 2 and 3, respectively.

3.1. Variable construction

For bonds included in our sample, we construct composite credit ratings by taking the mean of a bond's credit rating, where available, from Moody's, Standard & Poor's, and Fitch, and then rounding down. Furthermore, we also aggregate the composite credit ratings into credit rating buckets.

Bonds' sector categories are based on the Global Industry Classical Standard codes and used to analyze green bonds by industry. We tailor our sector categories to the major segments of the green bond market. Our green bond sector categories are alternate energy, banks, electric utilities and fossil fuels, industry and materials, nonbank financials, real estate, transportation, and others.[20]

Using bond-level data on the use of proceeds, we construct a refinancing variable equal to one if some portion of the bond was used to refinance an existing liability. Otherwise, this variable has a value of zero. We are able to construct this variable for 706 green bonds and 40,794 conventional bonds in our sample. Within this subset, 200 (28%) of the green bonds are used for refinancing, compared to 4,028 (10%) of conventional bonds.

3.1.1. Green bond variables

Additionally, we construct a set of variables primarily for the green bonds in our sample. In addition to the Bloomberg Back Office data, we work with supplementary data sources to construct some of these variables.

We collect each green bond's oversubscription ratio by combining data from Refinitiv's (2021) IFR and Bloomberg Finance LP (2021), where data are available.[21] A bond is oversubscribed if the investment bank underwriting the bond issuance receives excess orders relative to the actual amount of debt being issued. For a given bond, we thus calculate oversubscription as the ratio of the notional amount of orders to the actual amount of debt issued. Readily available data on investment bank order books for the corporate bond market is limited, but we are able to construct this variable for 481 of our 1,169 green bonds (41%). High oversubscription ratios can indicate strong demand for a bond issuance relative to supply. Compared to the typical demand seen for conventional bonds, green bonds have been well-received by investors, with some issues being several times oversubscribed, a fact that is often noted in the financial press.

We also construct a variable reflecting each bond's adherence to the GBP. At a minimum, green bonds must adhere to the first component of the GBP, *Use of Proceeds*, to be identified as a green bond by Bloomberg. The remaining three components of the GBP are optional and reflect the

[20] Transportation firms include firms in the transportation and auto manufacturing industry groups. Firms in the oil and gas sectors issue very few green bonds, so they are included with electric utilities. Table A2 in the Appendix provides a more detailed breakdown of the green bond sample by industry.

[21] In an earlier version of this paper, these data were primarily obtained from the textual bond note variable that contains additional information—in some cases, the oversubscription ratio—on the bond issue.

quality of the green bond's governance.[22] For each green bond, our GBP variable equals zero if a green bond is not fully aligned with the Green Bond Principles and one if the bond is fully aligned. We also construct a variable reflecting if the green bond was subject to a pre-issuance external review by a third party. While the external review is not a component of the GBP, it is considered "*recommended.*"

Finally, we construct a triple index inclusion indicator variable that is equal to one if a green bond was a constituent of three major green bond indices in our sample: the ICE Green Bond Index (542 bonds in the sample), the Solactive Green Bond Index (505), and the JP Morgan ESG Green Bond Index (436).[23] The triple index inclusion variable equals one for 365 green bonds in our sample.

3.2. Empirical approach

The full sample regressions in this paper are similar to those of Baker et al. (2022) in that we estimate a fixed-effects regression across an unbalanced panel of corporate bonds, with a given bond's yield spread at issuance as the dependent variable. The granularity of our data allows us to account for potential nonlinearities as well as issuer- and bond-specific time variation in the regressions. In the matched sample robustness tests (Flammer, 2021), we use similar multivariate regressions to control for differences in inexactly matched covariates. We provide a more detailed discussion of the matched sample regression specifications in Subsection 4.2.

We chose a fixed-effects regression approach as our primary empirical strategy over a matching approach for the following reason[24]: The matching approach either requires a triplet of bonds from the same issuer (one green bond and two comparable conventional bonds that are used to interpolate a localized conventional yield curve) or a set of comparable bonds that produce a balanced sample and allows one to account for the confounding bias from inexactly matched bond characteristics. In general, matching drastically reduces the number of green bonds that can be investigated and biases the sample to issuers with strong capital market access and the ability to issue comparable bonds frequently. Issuers that cannot frequently issue comparable bonds, such as small- and medium-sized enterprises (SMEs) and issuers in emerging markets, will be underrepresented in a matching approach.

In our regression approach, we compare the borrowing costs of green and conventional bonds using an indicator variable that flags green bonds while holding other factors constant. Our empirical baseline model is as follows:

$$\text{Yield spread}_{i,f} = \alpha\,\text{Green}_i + \beta\,\text{Controls}^T_{i,c,t} + \mu^T_{i,c,m,f} + \epsilon_{i,f} \tag{1}$$

for bond i of ultimate parent company f issued in currency region c on date t in year-month m. In Equation (1), the key variable of interest is the indicator variable Green_i, which takes the value of one if bond i is a green bond. The coefficient on this indicator variable, α, captures the average difference in primary market yield spreads for green and conventional bonds, holding other factors constant. A negative α indicates a lower yield spread for green bonds and, therefore, a borrowing cost advantage over conventional bonds. The vector $\text{Controls}^T_{i,c,t}$ contains bond- and macro-level controls, observed for bond i in currency region c on issue date t. The vector $\mu^T_{i,c,m,f}$ contains bond-, currency-, firm-, and time- fixed effects for bond i, currency region c, ultimate parent f, and year-month m, as well as an interaction of year-month and currency region, $m \times c$. Standard errors are clustered on the issuer ultimate parent and year-month levels and, thus, are robust to both cross-sectional dependence and serial correlation.

Specifically, our vector of controls includes the following variables. We have two bond-level numeric variables: the log years to maturity and the log notional amount issued. We further control for multiple variables that capture the general condition of credit markets in different currencies: The level, slope, and curvature of the sovereign yield curve for a given bond's currency region, calculated between 1 and 10 years to maturity using a principal component analysis, and the 30-day realized volatility of the 10-year sovereign bond for a given bond's currency region. To add a dimension for aggregate credit risk, we control for the option-adjusted spread of the ICE BofA global investment-grade corporate bond index (G0BC) and the difference in the OAS between the ICE BofA global high-yield corporate bond index (HW00) and the investment-grade corporate bond index.

In addition to the numeric controls, we include several fixed effects to capture potential nonlinearities: the issuer ultimate parent, the bond credit rating, seniority, whether a bond is callable, whether a bond is putable, whether a bond is sinkable, the year-month of issuance, and the currency of issuance interacted with year-month, as well as country of issuance interacted with an indicator variable marking SOEs.

Lastly, to control for potential bond-specific time variation in maturity risk, credit risk, and call-optionality of bonds, we allow for interactions of the numeric controls and the fixed effect. That is, we include interactions between the slope of the yield curve and log maturity; the bonds' rating bucket and the high-yield minus investment-grade index to allow rating buckets to load on aggregate credit risk; and whether a bond is callable and the level, slope, curvature, and realized volatility of the sovereign yield curve.[25] To control for issuer-specific time variation, we include time-varying issuer fixed effects, such as issuer ultimate parent interacted with annual and year-quarter fixed effects, respectively.

4. Empirical results

In this section, we present our empirical results. We start with a baseline estimate and robustness tests of the greenium at issuance. We then explore potential drivers behind the greenium. We investigate the timing of the greenium's emergence and the impact of proxies for demand pressure (oversubscription and bond index inclusion). We also differentiate the greenium by currency, whether or not corporations are local or foreign currency issuers, and whether the greenium relates to the "greenness" of the bonds as well as to specific bond- and firm-level characteristics.

4.1. Baseline regression results

Table 2 details the derivation of our greenium estimates by systematically introducing the control variables outlined in Subsection 3.2 into regression specification (1). Starting with a simple univariate regression of yield spreads on the green bond indicator variable in Column (1), we do not yet observe a negative or significant coefficient estimate. However, upon incorporating time-varying currency and issuer fixed effects (Columns (2) and (3)), the coefficient on the green bond indicator

[22] The remaining three components of the GBP are (1) *Process for Evaluation and Selection*, which indicates if the issuer has clearly communicated the eligibility, objectives, and potential risks of the bond's green projects; (2) *Management of Proceeds*, which indicates if the green bond's proceeds are transferred to a sub-account or sub-portfolio or otherwise managed in an "*appropriate manner*"; and (3) *Reporting*, which indicates if the issuer produces an annual report until the bond's proceeds are fully allocated. See International Capital Markets Association (2021).

[23] The JP Morgan index was launched in November 2020. Before November 2020, we rely on JP Morgan's simulated index constituents, which reflects what the index constituents would have been if the index methodology was retroactively applied to historical data.

[24] For applications of these two empirical approaches in the U.S. green municipal bond market, see Baker et al. (2022) and Larcker and Watts (2020).

[25] While of limited concern on a bond's issuance day, we control for interactions between the yield curve and whether a bond is callable following Gilchrist and Zakrajšek (2012).

variable becomes significantly negative.[26] The negative regression coefficients suggest the presence of a greenium, while holding the time, issuer, and currency dimension constant. Additionally, the increases in adjusted R-squared in Columns (2) and (3) indicate that much of the variation in yield spreads at issuance can be captured by controlling for time and issuer fixed effects already.

We observed little change in the coefficients estimates, with a greenium of around 11 basis points, once we add bond credit rating fixed effects in Column (3) as well as seniority, bond features (callable, putable, and sinkable), and industry and SOE times country fixed effects in Column (4). Lastly, adding bond-level numeric controls, controls for aggregate credit risk, and controls for interest rate conditions in Columns (6) to (8), respectively, the coefficient estimates on the green bond indicator variable are little changed, yielding a greenium of 11.4 basis points in the most rigorous specification in Column (8).

The controls display the expected theoretical and typical empirical relationship with bonds' yield spreads. For example, consistent with theory and upward-sloping spread-over-maturity curves, bonds' time-to-maturity significantly widens yield spreads. Broadly in line with estimates in Gilchrist and Zakrajšek (2012), increases in the level of interest rates and a steepening of a currency region's sovereign term structure lead to narrower yield spreads. Lastly, our measures of aggregate credit risk significantly widen bond spreads, particularly for bonds with lower credit quality.[27]

Next, we present our baseline results on the greenium and the green halo in Table 2b while introducing time-varying issuer fixed effects to address the concern that omitted time-varying issuer characteristics may drive our results. In Columns (1) and (2), we continue to find negative and highly significant coefficients on the green bond indicator variable. In comparison to the specification without time-varying issuer fixed effects in Table 2a Column (8), in which we estimate a greenium of 11.4 basis points, controlling for time-varying issuer fixed effects at the annual (Column (1)) or quarterly (Column (2)) frequency marginally decreases the greenium to about 8.6 basis points and 9.2 basis points, respectively.

We select the regression specification in Column (1), which uses annual time-varying issuer fixed effects, as our baseline estimate of the greenium. This choice is motivated by the following consideration: It is important that our regressions account for issuer-specific time-variation, but there is a trade-off between the granularity of the fixed effects and the sample size due to our elimination of singleton observations.[28] As shown in Column (2), increasing the granularity of the time-varying issuer fixed effects to the quarterly frequency does not meaningfully impact the regression estimate, but it causes a sharp drop in our sample size (see Fig. A1a in the Appendix for the impact of time-varying fixed effects on green bond sample size). This reduction in the sample impacts the bonds issued by smaller issuers, which are more likely to issue only a single bond in a given quarter and, thus, be counted as a singleton.

Taking Column (1) as our baseline regression, the green bonds in our sample receive an average yield spread at issuance of about 8 basis points lower relative to conventional bonds. The estimate suggests that, on average, corporate bond issuers pay a lower interest rate on their green bonds, holding other factors constant. Relative to the average yield spread in the corresponding regression sample, this reflects a roughly 7% decrease in the borrowing cost to the issuer. This borrowing cost advantage is economically meaningful, but at the same time, it is reduced by green bonds' compliance costs (certification, monitoring, and reporting) and a longer issuance process, particularly for complex green projects and small or first-time issuers.[29] However, the issuance costs may decrease as firms issue multiple green bonds, which is when an average 7% borrowing cost advantage may start to accumulate.

We next test whether green bond issuers receive a borrowing cost advantage across all of their bonds, both green and conventional.[30] We construct an indicator variable equal to one for a given bond if its issuer has previously issued a green bond (or if the given bond is the issuer's first green bond). Without time-varying issuer fixed effects, we find a small, negative coefficient of about 5 basis points on the green issuer indicator that is significant at the 5% significance level (Column (3)). Controlling for time-varying issuer fixed effects, the coefficient on the green issuer variable becomes neither economically meaningful nor statistically significant (Column (4)). Green issuers, relative to conventional issuers, thus do not appear to capture a robust overall borrowing cost advantage across all of their bonds. Next, in Columns (5) and (6), we test if the borrowing cost advantage of the greenium is distinct from potential benefits for green issuers' other bonds. While the greenium coefficient remains highly significant at 9 basis points and 8 basis points, the coefficient on the green issuer indicator variable is not statistically significant in both specifications. In combination, the results in Columns (3) to (6) suggest that the greenium is a separate phenomenon from any potential green halo benefit to the remaining bonds of green bond issuers.

4.2. Robustness in subsamples and matched samples

Next, we conduct two sets of robustness tests on the baseline greenium estimates that address concerns about sample balance (129,043 conventional bonds versus 1,169 green bonds) and differences in issuer characteristics (across versus within issuer estimates). First, we re-estimate our baseline fixed-effects regression model in three progressively more balanced subsamples of the full dataset. Second, we adopt a careful nearest-neighbor matching methodology among primary market bond issues by extending the bond matching approach in Flammer (2021).

While the sample's skewness towards conventional bonds partly reflects the still small share of green bonds in firms' total bond issuance, we address potential concerns on sample balance by increasing the degree to which the control group of conventional bonds represents the subset of green bonds. Specifically, in Table 3, we sequentially align green and conventional bonds based on bond and issuer characteristics.

Because exact covariate matching on currency and credit rating bucket in the matching exercise limits the matched samples to U.S. dollar- and euro-denominated bonds with a credit rating, we first re-estimate our fixed-effects regression model in a subsample of rated bonds denominated in the euro or U.S. dollar (that is, 34,262 conventional bonds to 587 green bonds such that green bonds represent about

[26] Without the interaction of currency times year-month fixed effects, the baseline greenium estimates remain qualitatively and quantitatively similar. Nevertheless, given the multi-currency panel dataset spanning 2014 to 2021, accounting for heterogeneity in time variation across currency regions seems prudent.

[27] In unreported results, we also find that other sources of bond risk, such as the bond credit rating fixed effects (higher credit risk monotonically widens yield spreads) and the seniority fixed effects (higher seniority is associated with tighter yield spreads), show the appropriate signs when included in the regression as indicator variables.

[28] See Correia (2015) for discussion of singleton observations.

[29] In unreported results that go beyond the scope of this analysis, we find that, on average, underwriters charge lower fees for underwriting green bonds than conventional bonds. This might reflect underwriters' incentives to establish a foothold in the nascent market for issuing green bonds.

[30] This potential benefit to a firm's overall cost of capital has been called the "green halo." Most empirical research on the green halo—see, for example, Flammer (2021), Baulkaran (2019), and Tang and Zhang (2020)—has focused on the effects of the announcement of a first green bond issue on the firm's equity and secondary bond yields. These announcement returns are captured by stockholders and bondholders but do not necessarily impact the firm's actual cost of capital beyond a short-term window. In a broader context, green bonds may, thus, serve as a solution to a signaling problem about firms' environmental credentials (Daubanes et al., 2021).

Table 2
Greenium baseline regressions (full sample).

(a) Greenium baseline regressions and impact of control variables

	Yield spread							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Green	12.56	-21.88*	-10.47***	-13.33***	-11.38***	-11.86***	-11.74***	-11.43***
	(12.61)	(11.60)	(2.858)	(2.849)	(2.601)	(2.458)	(2.384)	(2.360)
Size						-0.0714	0.0477	0.0148
						(0.738)	(0.727)	(0.719)
Maturity						17.69***	17.85***	18.02***
						(1.786)	(1.777)	(1.745)
IG index spread							0.242***	0.390***
							(0.0822)	(0.0862)
A × HY-IG index spread							0.0595	-0.0363
							(0.0444)	(0.0454)
AA × HY-IG index spread							0.0551	-0.0329
							(0.0616)	(0.0615)
AAA × HY-IG index spread							-0.0457	-0.133**
							(0.0513)	(0.0518)
B × HY-IG index spread							0.458***	0.345***
							(0.0711)	(0.0717)
BB × HY-IG index spread							0.449***	0.339***
							(0.0669)	(0.0678)
BBB × HY-IG index spread							0.197***	0.0941
							(0.0578)	(0.0585)
C × HY-IG index spread							0.312**	0.191
							(0.129)	(0.127)
Not rated × HY-IG index spread							-0.00197	-0.0836*
							(0.0475)	(0.0492)
Level								-0.152***
								(0.0276)
Slope								-0.0314
								(0.0745)
Slope × Maturity								-0.0729***
								(0.0178)
Curvature								0.114
								(0.118)
Realized vol.								0.271
								(0.273)
Level × Callable								-0.0381***
								(0.00992)
Slope × Callable								-0.0310
								(0.0284)
Curvature × Callable								-0.161
								(0.132)
Realized vol. × Callable								-0.491
								(0.409)
Currency × Month FE		✓	✓	✓	✓	✓	✓	✓
Firm FE			✓	✓	✓	✓	✓	✓
Rating FE				✓	✓	✓	✓	✓
Seniority FE					✓	✓	✓	✓
Bond Option FEs					✓	✓	✓	✓
Industry & SOE FEs					✓	✓	✓	✓
Observations	130,212	130,084	126,373	126,373	126373	126,373	126,373	126,373
R^2	0.000	0.226	0.754	0.769	0.779	0.781	0.783	0.784
Adjusted R^2	0.000	0.214	0.731	0.747	0.757	0.760	0.762	0.763
Relative Greenium	9.1%	15.9%	7.8%	10.0%	8.5%	8.9%	8.8%	8.6%

Regressions of corporate yield spreads at issuance on the *Green* bond indicator variable, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variable, *Green*, takes the value of one for green bonds and zero otherwise. All other control variables and fixed effects are detailed in Section 3. *Relative Greenium* represents the absolute value of the ratio of the coefficient estimate on the green bond indicator variable to the average yield spread in the corresponding regression sample. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

(*continued on next page*)

Table 2 (*continued*)
(b) Greenium and green halo baseline regressions

	Yield spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Green	-8.623***	-9.166***			-9.735***	-8.646***
	(2.530)	(2.523)			(2.656)	(2.737)
Green issuer			-5.065**	-2.043	-4.157	0.0976
			(2.545)	(3.969)	(2.629)	(4.185)
Currency × Month FE	✓	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Seniority FE	✓	✓	✓	✓	✓	✓
Bond Option FEs	✓	✓	✓	✓	✓	✓
Industry & SOE FEs	✓	✓	✓	✓	✓	✓
Bond & Market Controls	✓	✓	✓	✓	✓	✓
Firm × Year FE	✓			✓		✓
Firm × Quarter FE		✓				
Observations	114,879	102,124	126,373	114,879	126,373	114,879
R^2	0.842	0.872	0.783	0.842	0.783	0.842
Adjusted R^2	0.809	0.833	0.762	0.809	0.762	0.809
Relative Greenium	6.9%	7.7%	-	-	7.3%	6.8%

Regressions of corporate yield spreads at issuance on the green bond and green issuer indicator variables, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variables are defined as follows: *Green* takes the value of one for green bonds and zero otherwise. *Green issuer* is equal to one for a given bond if its issuer has previously issued a green bond (or if the given bond is the issuer's first green bond). All other control variables and fixed effects are detailed in Section 3. *Relative Greenium* represents the absolute value of the ratio of the coefficient estimate on the green bond indicator variable to the average yield spread in the corresponding regression sample. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Table 3
Greenium robustness regressions (subsamples).

	Yield spread		
	(1)	(2)	(3)
Green	-8.148***	-7.565***	-5.744**
	(2.384)	(2.255)	(2.465)
Currency × Month FE	✓	✓	✓
Rating FE	✓	✓	✓
Seniority FE	✓	✓	✓
Bond Option FEs	✓	✓	✓
Industry & SOE FEs	✓	✓	✓
Bond & Market Controls	✓	✓	✓
Firm × Year FE	✓	✓	✓
Observations	34,849	11,457	6,810
R^2	0.944	0.856	0.896
Adjusted R^2	0.930	0.832	0.877
Relative Greenium	4.6%	6.9%	5.3%

Regressions of corporate yield spreads at issuance on the green bond indicator variable, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). Regressions are estimated on three sub-samples of the full dataset, which contain (1) rated, euro- or U.S. dollar-denominated bonds, (2) rated, euro- or U.S. dollar-denominated bonds from green issuers, and (3) rated, euro- or U.S. dollar-denominated bonds from green issuers for which we successfully match bonds. The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variable, *Green*, takes the value of one for green bonds and zero otherwise. All other control variables and fixed effects are detailed in Section 3. *Relative Greenium* represents the absolute value of the ratio of the coefficient estimate on the green bond indicator variable to the average yield spread in the corresponding regression sample. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

1.7% of the subsample). Compared to the full sample greenium estimate of 8.6 basis points, we find a nearly identical coefficient estimate at negative 8.1 basis points in Column (1).

When we further restrict the subsample to rated U.S. dollar- or euro-denominated bonds from green bond issuers in Column (2) (that is, 10,870 conventional bonds to 587 green bonds such that green bonds represent about 5.1% of the subsample), we find a largely comparable greenium estimate of 7.5 basis points. Lastly, in an even more restrictive subsample that contains only rated U.S. dollar- or euro-denominated bonds from green bond issuers for which we successfully match bonds in the subsequent matching exercise (that is, 6,414 conventional bonds to 396 green bonds such that green bonds represent about 5.8% of the subsample), we find an only marginally lower average greenium of 5.7 basis points.

Overall, the subsample regressions produce qualitatively and quantitatively similar greeniums, confirming a robust borrowing cost advantage in more balanced subsets of the data. Furthermore, these estimates continue to reflect economically meaningful discounts of 4% to 7%, respectively, in firms' borrowing costs relative to the average yield spread in the corresponding regression samples.

We adopt a nearest-neighbor matching methodology to address the concern that yield spread differences between green and conventional bonds might arise from uncontrolled factors in our fixed-effects regression specifications (Larcker and Watts, 2020). Specifically, in the six months before and after a green bond's issuance, we match the green bond to its most comparable conventional bond from the same issuer. The matching procedure is done in three steps: First, we align bonds on four sources of bond price risk by requiring that the credit rating bucket, currency, seniority, coupon type, and callable bond feature of the bonds be the same. Within a rating bucket (for example, credit ratings AA-, AA, and AA+), bonds' credit ratings may not differ by more than one rating notch.[31,32] Second, we pick the nearest one-to-one neighbor

[31] Most bonds in our matched samples have identical credit ratings (about 93% of the matched bonds). This is partly the result of our Mahalanobis distance measure penalizing rating deviations when determining the closest neighbors. Our results remain qualitatively and quantitatively similar when we match on bonds' exact credit ratings.

[32] About 45% of bonds in the full sample show one-notch rating differences when bonds are rated by more than one rating agency. Thus, one may argue that a one-notch rating difference within the same rating bucket only marginally changes a bond's underlying credit risk. Ultimately, we opt to allow one-notch rating differences within the same rating bucket because it preserves precision through a higher number of matches while not materially impacting the bias in the matched sample. This flexibility becomes relevant when one can trade off a potential one-notch rating difference against smaller differences in other rel-

Table 4
Covariate balance in matched samples.

One-to-one Matched Bond Sample								
	Green			Conventional			Covariate Balance	
	Observations	Mean	S.D.	Observations	Mean	S.D.	Standardized Diff.	Variance Ratio
Yield spread (bp)	254	184.41	163.30	259	186.44	163.37	0.01	1.00
Size	254	20.11	0.75	259	20.11	0.77	0.00	0.95
Maturity	254	1.90	0.47	259	1.87	0.50	0.06	0.88
Coupon (%)	254	2.66	2.11	259	2.67	2.13	0.00	0.99
Rating	254	8.28	3.80	259	8.29	3.80	0.00	1.00
Three-to-one Matched Bond Sample								
	Green			Conventional			Covariate Balance	
	Observations	Mean	S.D.	Observations	Mean	S.D.	Standardized Diff.	Variance Ratio
Yield spread (bp)	254	170.04	156.66	361	171.74	157.26	0.01	0.99
Size	254	20.11	0.82	361	20.11	0.84	0.00	0.96
Maturity	254	1.88	0.46	361	1.84	0.50	0.08	0.84
Coupon (%)	254	2.46	2.06	361	2.45	2.08	0.00	0.99
Rating	254	7.85	3.79	361	7.85	3.80	0.00	0.99

Covariate balance statistics for the one-to-one and three-to-one matched bond samples, separating green and conventional bonds. Bond types are compared across the following metrics: *Yield spread (bp)* is bonds' yield spread at issuance, reported in basis points. *Size* is the logarithm of the bond's notional amount issued. *Maturity* represents the logarithm of bonds' time-to-maturity (in years) on the issuance day. *Coupon (%)* is bonds' coupon (for fixed coupon bonds only) and reported in %. *Rating* represents a bond's credit rating, ranging from AAA = 1 to C = 21. We define a bond's credit rating as the mean of credit ratings, where available, from Moody's, Standard & Poor's, and Fitch, and then rounding down. The last two columns assess covariate balance based on the absolute value of the standardized difference in means and the variance ratio (Imbens and Wooldridge, 2009; Austin, 2009). Differences between the upper and lower panels reflect sample composition and weighting. In cases of (up to) three matches, bonds' weights are determined based on Mahalanobis distances.

Table 5
Greenium robustness regressions (matched samples).

	One-to-one Matched Sample				Three-to-one Matched Sample			
	Yield spread				Yield spread			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Green	-2.023***	-2.574**	-3.154***	-2.774***	-1.696	-2.733**	-3.795***	-2.941**
	(0.652)	(1.094)	(0.843)	(0.849)	(1.314)	(1.247)	(1.238)	(1.386)
Bond Group FE		✓	✓	✓		✓	✓	✓
Month FE		✓	✓	✓		✓	✓	✓
Rating FE			✓	✓			✓	✓
Bond Option FE			✓	✓			✓	✓
Currency FE			✓	✓			✓	✓
Bond & Market Controls				✓				✓
Observations	513	499	499	499	615	608	608	608
R^2	0.00	0.99	0.99	0.99	0.000	0.99	0.99	0.99
Adjusted R^2	0.00	0.98	0.99	0.99	0.00	0.99	0.99	0.99
Relative Greenium	1.1%	1.4%	1.7%	1.5%	1.0%	1.6%	2.2%	1.7%

Regressions of corporate yield spreads at issuance on the green bond indicator variable, using matched bond samples that span the period 2014 through 2021 (Subsection 4.2 details the matching methodology). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variable, *Green*, takes the value of one for green bonds and zero otherwise. All other control variables and fixed effects are detailed in Section 3 as well as Subsection 4.2. *Relative Greenium* represents the absolute value of the ratio of the coefficient estimate on the green bond indicator variable to the average yield spread in the corresponding regression sample. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

using the Mahalanobis distance, which we compute based on four characteristics: (i) log (issuance amount), (ii) time-to-maturity (in years), (iii) coupon, and (iv) the exact credit rating within the rating bucket.[33] Third, after steps 1 and 2, the matched sample shows significant covariate imbalance in bonds' offering amounts and time-to-maturity using common measures of covariate balance (see, for example, Imbens and Wooldridge (2009)). Because of this imbalance, we place calipers on the two covariates to ensure a satisfactory balance.[34]

To examine the sensitivity of the *one-to-one* matching procedure to the inherent bias-variance trade-off, we repeat steps 1 to 3 and also select the nearest *three-to-one* neighbors, which pairs (up to) three conventional bonds with each green bond.[35] Importantly, increasing the

evant risk factors (maturity, size, or coupon) to determine the closest neighbor using the Mahalanobis distance measure.

[33] Compared to Flammer (2021), we do not include the number of days in between the green and conventional bond issuance. Instead, we require that bonds be issued within a 12-month window around the green bond issuance date, and then we control for changes in interest rate and credit risk conditions in matched sample regressions.

[34] We require that matched bonds deviate by not more than ±2.5% from the matched green bond's issue size (75th percentile of matches) and allow years-to-maturity to deviate by up to ±3 years (50th percentile of matches).

[35] Increasing the potential number of matches can improve precision by preventing too many eligible conventional bonds from being dropped from the

number of matches does not materially worsen covariate balance, but we gain some precision in the matched sample regressions due to a larger effective sample size. We end up with a matched one-to-one sample with 254 green and 259 conventional bonds and a matched three-to-one sample with 254 green and 361 conventional bonds that reflect bonds from 156 green bond issuers.[36] Absolute values in standardized differences in means as well as variance ratios for key bond characteristics – namely, the yield spread, size, years-to-maturity, coupon, and credit rating – are reported in Table 4 and fall well within permissible bounds.[37] Nevertheless, by pairing up to three conventional bonds with each green bond, we find that covariate balance in bonds' years-to-maturity marginally declines, as the standardized difference in means increases from 0.06 to 0.08, while the variance ratio decreases from 0.88 to 0.84.

The matched samples still represent unbalanced panels of firms' bonds, in which (i) firms may have issued multiple green and conventional bonds at different points in time, and (ii) bonds from multiple firms may have been issued in the same year and month. This introduces the potential for cross-sectional dependence and serial correlation in bonds' yield spreads. Furthermore, even though bonds have been matched as closely as possible, there may be relevant differences in inexactly matched covariates (for example, maturity) and differences in interest rates and credit risk conditions at the bonds' time of issuance. In addition to a simple univariate comparison of average yield spreads, we therefore also perform a more appropriate matched sample regression analysis based on the following specification:

$$\text{Yield spread}_{i,f} = \alpha\,\text{Green}_i + \beta\,\text{Controls}^T_{i,c,t} + \mu^T_{r,c,g} + \tau_m + \epsilon_{i,f} \tag{2}$$

for bond i of ultimate parent company f issued in currency region c on date t in year-month m. The vector $\text{Controls}^T_{i,c,t}$ controls for observed differences in matched bonds' time-to-maturity and size, for differences in a currency region's interest rate conditions (level, slope, curvature) and their interactions in case bonds are callable, as well as aggregate credit risk conditions (global investment-grade and investment-grade minus high-yield bond indexes' yield spreads). We further control for potential one-notch rating differences with a rating fixed effect, μ_r, and potential difference between U.S. dollar- and euro-denominated green bonds with a currency fixed effect, μ_c. These controls and fixed effects are largely identical to those in Equation (1). To account for unobserved matched group differences, we add the group fixed effect μ_g, which separately differentiates each matched bond group. With only one matched bond group per issuer, μ_g represents an issuer fixed effect. With multiple matched bond groups per issuer, μ_g works as a time-varying issuer fixed effect. Lastly, given the eight-year sample period and "treatment" across time (here, green bond issuances), we also account for unobserved time variation at the month-year level with τ_m. Standard errors are clustered on the issuer ultimate parent and year-month levels.

Table 5 presents the matched sample regressions results. Using a simple univariate regression of yield spreads on the green bond indicator variable, we find an average greenium of 2 basis points in the one-to-one matched sample (Column (1)) and 1.7 basis points in the three-to-one matched sample (Column (5)), respectively. The univariate greenium is highly significant in the one-to-one matched sample but insignificant in the three-to-one matched sample, which could result from the three-to-one matched sample's higher covariate imbalance and, thus, a confounding bias. In fact, once we control for bond differences, by adding bond and market controls in Columns (2) to (4) and (6) to (8), the average greenium estimate in both matched samples increases to a significant 3 basis points. While the matched sample estimates are smaller than the full sample estimates, they remain highly statistically significant and continue to reflect economically meaningful discounts of about 2 of the firms' borrowing costs relative to the average yield spread in the corresponding samples.

Overall, our estimates are consistent with a small but significant average greenium, even in the very restrictive matched sample regressions. Since our analysis is primarily focused on the sign and significance of the greenium and not an exact size, the robustness tests confirm that green bonds capture a borrowing cost advantage at issuance over their conventional counterparts.

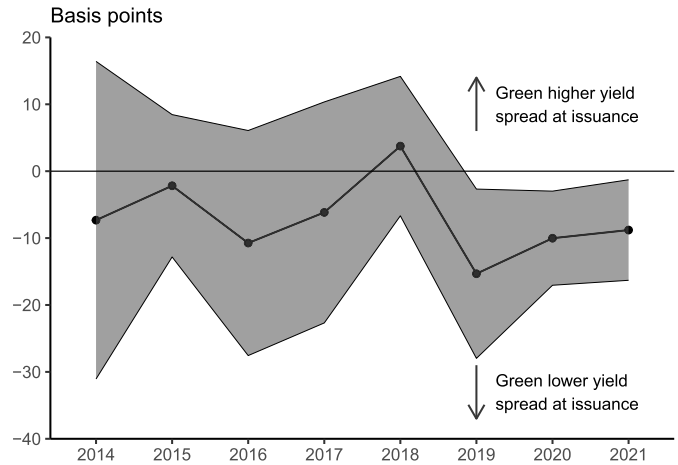

Fig. 3. Time series of corporate bond greenium at issuance. Regression estimates from Table A3 of the average annual yield spread difference between green and conventional corporate bonds, holding other factors constant. Negative values indicate green corporate bonds have a lower yield spread at issuance than conventional corporate bonds and, therefore, a borrowing cost advantage. The shaded ribbon indicates the 95% confidence interval.

4.3. Time-variation in the greenium

In Fig. 3, we use a similar regression specification to Equation (1) to construct a time series of the greenium at issuance at an annual frequency. We interact the green bond indicator variable with a yearly time fixed effect; the regression estimates are detailed in Table A3 in the Appendix. In Fig. 3, black dots reflect the greenium coefficient for each year, and the gray ribbon reflects a 95% confidence interval. We find that a statistically significant average greenium in our sample only emerges in 2019 at about 15 basis points and then tightens to about 10 basis points and 8 basis points in 2020 and 2021, respectively.

The emergence of the significant average greenium coincides with the release of the European Union Sustainable Finance Action Plan (EU SFAP) by the European Commission in March 2018. The plan reflected a broad commitment to reorient capital flows toward a more sustainable economy, with specific commitments to standardize and regulate green securities through regulations such as the EU Green Bond Standard. The plan included a regulation to mandate and standardize the sustainability disclosures of financial market participants (or SFDR) in an effort to increase transparency. Following the release of the action plan by the European Commission, the final version of the SFDR was eventually adopted by the European Parliament on April 18, 2019, and signed into law on November 27, 2019.

sample. This gain in precision typically diminishes after four matches as covariate imbalance and, thus, bias increases (see, for example, (Rosenbaum, 2020)). Therefore, we limit the matching ratio to three-to-one. When a green bond is matched with multiple conventional bonds, we weigh matches based on their Mahalanobis distances, so closer matches receive larger weights.

[36] A few green bonds have tied nearest neighbors. We do not break ties; instead, we weigh tied neighbors appropriately. Our results remain qualitatively and quantitatively similar when we break ties randomly.

[37] A standardized difference in means that exceeds 0.25 signals that the matched samples may be unbalanced (Imbens and Wooldridge, 2009). Similarly, variance ratios should be approximately 1 and lie within the range of [0.78,1.28] for our matched samples (the 2.5th and 97.5th percentiles of an F-distribution with degrees of freedom implied by the number of matched bond groups (Austin, 2009)).

Table 6
Greenium and EU regulation, by currency.

	Yield spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Green			-2.048	-4.326		
			(3.521)	(3.622)		
Green × Pre-SFDR	-2.048	-4.326				
	(3.521)	(3.622)				
Green × Post-SFDR	-14.71***	-10.25***	-12.67***	-5.922		
	(3.151)	(3.159)	(4.405)	(4.388)		
Green × EUR					-0.176	-1.163
					(3.192)	(3.110)
Green × EUR × Post-SFDR					-11.64***	-5.912
					(3.952)	(3.774)
Green × USD					-4.165	-7.868
					(6.473)	(6.727)
Green × USD × Post-SFDR					-15.68*	-8.523
					(8.269)	(8.211)
Currency × Month FE	✓	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Seniority FE	✓	✓	✓	✓	✓	✓
Bond Option FEs	✓	✓	✓	✓	✓	✓
Industry & SOE FEs	✓	✓	✓	✓	✓	✓
Bond & Market Controls	✓	✓	✓	✓	✓	✓
Firm × FE	✓		✓		✓	
Firm × Year FE		✓		✓		✓
Observations	126,373	114,879	126,373	114,879	126,373	114,879
R^2	0.784	0.843	0.784	0.843	0.784	0.843
Adjusted R^2	0.763	0.810	0.763	0.810	0.763	0.810

Regressions of corporate yield spreads at issuance on the green bond and EU regulation indicator variables, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variables are defined as follows: *Green* takes the value of one for green bonds and zero otherwise. *Pre-SFDR* equals one before April 18, 2019, and zero afterward. *Post-SFDR* equals zero before April 18, 2019, and one afterward. *EUR* and *USD* are indicator variables that differentiate euro- and U.S. dollar-denominated green bonds, respectively. All other control variables and fixed effects are detailed in Section 3. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

We test whether the EU regulation contributed to the emergence of the average greenium in 2019 by creating the indicator variable SFDR_t, which is equal to zero before and equal to one on and after April 18, 2019, the date the EU Parliament adopted the SFDR.[38] In Table 6 Column (1), we estimate the greenium in the pre- and post-SFDR periods. In line with our results from Fig. 3, we find a significantly negative greenium of about 15 basis points in the period following the EU regulation. In Column (2), we show that this post-SFDR greenium is little changed by the inclusion of annual time-varying issuer fixed effects, remaining significantly negative at about 10 basis points.

In Columns (3) and (4), we adopt a difference-in-differences specification, including the green bond indicator variable in combination with the post-SFDR interaction term. The coefficient on the interaction term should capture the response in the greenium following the regulatory change.[39] In Column (3), without controlling for time-varying issuer fixed effects, we still find a significantly negative coefficient on the interaction term. However, in Column (4), controlling for annual time-varying issuer fixed effects, the interaction term is no longer significant, suggesting that issuer-specific time-variation rather than just the regulatory change contributed to the significance of the greenium as of 2019.

In Columns (5) and (6), we differentiate the greenium into the pre- and post-SFDR periods with further segmentation by currency. Consistent with Columns (1) through (4), we find significant negative greenium coefficients for euro and U.S. dollar green bonds in Column (5). However, these coefficients become insignificant with the addition of time-varying issuer fixed effects in Column (6).

While a significant greenium emerged as of 2019, we cannot conclude empirically that the regulation caused its emergence. However, this is not to say that the EU SFAP and the SFDR have been unsupportive of the overall development of the green bond market and its strength in Europe.

4.4. Proxies for excess demand

We find a significant average greenium that emerges in 2019. In this subsection, we focus on the impact of supply and demand imbalances for green bonds and how demand pressure relative to a green bond's fundamentals could drive the greenium. We examine two proxies for excess demand that could contribute to the formation of the greenium: Oversubscription and bond index inclusion.

[38] Results remain qualitatively similar for November 27, 2019, the day the SFDR was signed into law.

[39] The baseline effect for the post-SFDR indicator is partialled out due to our year-month fixed effects in our regressions.

4.4.1. Oversubscription

When firms experience high demand for their debt, they can raise the price of the debt (lower the yield), supply more debt, or do both (Ma, 2019; Mota, 2020; Bolton et al., 2013; Baker and Wurgler, 2002). An elastic supply makes it difficult to econometrically identify excess demand for a firm's debt simply by regressing the quantity supplied on yield spreads, prompting Siani (2022) to use intra-day bond supply data on the issuance day to show that firms respond to high demand in conventional U.S. investment-grade bonds by increasing the quantity supplied. In our setting, the notional size of a green bond is constrained by the opportunities for green investment available to an issuer. If there is very strong demand for a given issuer's green bond, it is difficult to increase the amount of debt issued without degrading the "greenness" of the bond. Instead, it is more feasible for the issuer to capture the high demand with a lower bond yield instead of a larger supply of debt. Thus, we argue that oversubscription rates are a good proxy for excess demand in the case of green bonds, allowing us to capture how issuers can benefit from demand pressure at issuance.[40] Holding other factors constant, excessive demand from investors relative to the notional size of a green bond may allow an underwriter to lower the bond's yield on behalf of the issuer. Thus, we hypothesize that oversubscription rates are negatively correlated with yield spreads at issuance.

In the full sample, the oversubscription analysis is limited to green bonds because our data coverage for conventional bonds is poor. As a result, we can only include an interaction term between the green bond indicator variable and the logarithm of green bonds' oversubscription ratios in our regression model. Without a baseline effect, the limited yet informative hypothesis we can test is whether green bonds with higher oversubscription ratios show lower yield spreads relative to green bonds with lower oversubscription ratios. What we cannot test is whether green bonds with higher oversubscription ratios show lower spreads than conventional bonds with comparable oversubscription ratios.

Because of the data limitation, we also conduct an oversubscription analysis in the matched sample. Compared to the full sample, the considerably smaller number of observations allows for a targeted collection of oversubscription data for conventional bonds. While we still lose more than half of the bonds in the three-to-one matched sample due to missing data,[41] the matched sample allows us to study the effect of bonds' oversubscription ratios on yield spreads, with both a baseline effect and an interaction term, in a balanced group of bonds. We report the full sample regression results in Table 7a, the summary statistics on oversubscription rates in the matched sample in Table 7b, and the matched sample results in Table 7c.

The estimate in Table 7a Column (1) indicates that there is a negative and significant relationship between green bonds' oversubscription ratios and their yield spreads at issuance. In our estimation, a 1% increase in the oversubscription ratio is associated with a 0.065 basis point increase in the greenium. For the full sample's average log oversubscription ratio of 1.48, this indicates an average yield spread of about 9 basis points, consistent with our baseline greenium results in Table 2b Column (1). In Column (2), we continue the analysis by controlling for the green bond indicator. Despite some collinearity between the two variables, we find that the oversubscription effect remains weakly significant and negative, while the coefficient on the green bond indicator turns insignificant.

In Column (3), we explore a potential nonlinear relationship between oversubscription ratios and yield spreads by breaking the green bonds with oversubscription data into terciles we form on their oversubscription ratios. Only highly oversubscribed green bonds receive a statistically significant reduction in borrowing costs at about 17 basis points. In Column (4), we confirm this effect when controlling for the green bond indicator variable, finding statistically significant differences in the coefficients for low, medium, and high oversubscription ratios. In Columns (5) and (6), we refine our result by segmenting the green bonds with oversubscription data by currency. We find that oversubscription in euro-denominated green bonds leads to a statistically significant reduction in yield spreads; for dollar green bonds, the effect is insignificant.

In Table 7c, we first confirm that green bonds are, on average, more oversubscribed than their comparable conventional counterparts. In a simple univariate regression of the logarithm of bonds' oversubscription rates on the green bond indicator variable (Column (1)), we find that green bonds have a 0.12 higher average log oversubscription rate than otherwise similar conventional bonds from the same issuer. This is economically meaningful, as it reflects roughly 11% of the matched sample's average log oversubscription rate. Once we add bond and market controls in Column (2) (Subsection 4.2 details control variables and fixed effects for the matched sample), the average difference in oversubscription rates is roughly halved but remains highly statistically significant.

In Columns (3) to (8), we resume the study of oversubscription ratios on yield spreads. Column (3) confirms that the subset of matched bonds with oversubscription data yields a significant greenium. Relative to the estimate in Column (8) of Table 5, the coefficient estimate doubles, now reading 6.9 basis points, which could suggest that green bonds for which underwriters report oversubscription data are more oversubscribed and, thus, able to capture a larger borrowing cost advantage. In Column (4), we confirm that higher investor demand, independent of bond type, significantly lowers yield spreads, while the coefficient on the green bond indicator variable is little changed. Once we differentiate oversubscription rates for green and conventional bonds in Column (5), we find a slightly larger and more significant coefficient for conventional bonds. However, in untabulated results, this difference in coefficients of 3.36 is statistically insignificant (t-stat: 1.10, p-value: 0.28), suggesting that the association between investor demand and lower yield spreads at issuance is not inherently different between bond groups. The greenium estimate increases to 9.8 basis points, which shows that investor demand cannot decisively lower the size and significance of the green borrowing cost advantage in this regression specification.

However, because institutional bond investors often reach for yield, regressions of investor demand on yield spreads can produce counterintuitive positive relationships (see, for example, Becker and Ivashina (2015); Choi and Kronlund (2018)). For highly sought-after green bonds, investors' reach for yield could thus bias the estimate on oversubscription ratios upward in Column (5). Following the approach in the contemporaneous research by Wang and Wu (2023), we add bonds' initial yield spread to the regression specification to also account for their attractiveness to investors at issuance.[42] As shown in Column (6), we still find a negative and significant relation between investor demand and yield spreads alongside a smaller but still significant greenium. When we differentiate investor demand by bond group (Column (7)), the coefficient on the green bond indicator variable turns insignificant, suggesting that once bonds' attractiveness at issuance is considered, excess demand fully explains the borrowing cost advantage of

[40] Anecdotally, market participants have reported that green corporate bonds can be heavily oversubscribed, indicating excess demand for a bond in the primary market (see, for example, Climate Bonds Initiative (2021).)

[41] Due to the limited data coverage, we only conduct our analysis in the larger three-to-one matched sample. Moreover, oversubscription data for high-yield bonds are particularly scarce, so we focus the analysis only on investment-grade bonds. Whenever we lose bonds to missing data in matched bond groups, we re-weight bonds accordingly.

[42] The initial yield spread represents the spread that underwriters propose to investors at the beginning of the book-building process (as opposed to the yield spread once the bond is priced on issuance day).

Table 7
Greenium and bond oversubscription.

(a) Green bond oversubscription and yield spreads (full sample)

	Yield spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Green × Oversubscription	-6.479***	-11.67*				
	(1.988)	(6.041)				
Green		8.172		-18.66***		10.74
		(8.078)		(5.455)		(8.557)
Green × Low oversubscription bucket			-5.253*	13.40**		
			(3.022)	(6.166)		
Green × Medium oversubscription bucket			-3.732	14.92**		
			(4.545)	(6.112)		
Green × High oversubscription bucket			-18.66***			
			(5.455)			
Green × EUR × Oversubscription					-8.746***	-15.85**
					(2.913)	(7.300)
Green × USD × Oversubscription					-3.780	-10.27*
					(2.846)	(5.730)
Currency × Month FE	✓	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Seniority FE	✓	✓	✓	✓	✓	✓
Bond Option FEs	✓	✓	✓	✓	✓	✓
Industry & SOE FEs	✓	✓	✓	✓	✓	✓
Bond & Market Controls	✓	✓	✓	✓	✓	✓
Firm × Year FE	✓	✓	✓	✓	✓	✓
Observations	114,168	114,168	114,168	114,168	114,168	114,168
R^2	0.843	0.843	0.843	0.843	0.843	0.843
Adjusted R^2	0.809	0.809	0.809	0.809	0.809	0.809

Regressions of corporate yield spreads at issuance on the green bond indicator variable and bond oversubscription variables, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variables are defined as follows: *Green* takes the value of one for green bonds and zero otherwise. *Oversubscription* is the ratio of the notional amount of orders to the actual amount of issued debt. *Low/medium/high oversubscription bucket* represent terciles formed on bonds' oversubscription ratios. *EUR* and *USD* are indicator variables that differentiate euro- and U.S. dollar-denominated green bonds. All other control variables and fixed effects are detailed in Section 3. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

(b) Three-to-one matched sample oversubscription statistics

	Observations	Mean	S.D.	25th	Median	75th
Green bonds						
Oversubscription	105	3.66	1.94	2.20	3.10	4.40
Initial Yield Spread	105	131.60	62.91	90.00	124.85	162.00
Conventional bonds						
Oversubscription	153	3.31	1.88	1.88	2.67	4.20
Initial Yield Spread	153	131.24	66.76	95.00	117.50	160.00

Summary statistics for bonds with reported oversubscription data in the three-to-one matched sample, separating summary statistics for green and conventional bonds. *Oversubscription* represents bonds' oversubscription ratio, defined as the ratio of the notional amount of bond orders to the actual amount of debt that is issued. *Initial Yield Spread* represents bonds' yield spread that underwriters propose to investors at the beginning of the book-building process (as opposed to bonds' yield spread once the bond is priced on issuance day).

(*continued on next page*)

green bonds. While the difference of 0.20 between the two coefficients is not statistically significant (t-stat: 0.11, p-value: 0.92), the, on average, higher oversubscription rates of green bonds point to a more pronounced reduction in yield spreads of close to 2 basis points (that is, -18.13 * 1.06 = -19.21 versus -17.93 * 1.17 = -20.98). Lastly, using a nonlinear measure of oversubscription rates, we confirm that the most oversubscribed bonds primarily drive this effect. Overall, the matched sample results confirm our existing findings and improve our understanding of the mechanism at hand since we can now also control for the baseline effect of oversubscription ratios.

Table 7 (*continued*)

(c) Greenium and oversubscription (three-to-one matched sample)

	Oversubscription		Yield spread					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Green	0.118***	0.0688***	-6.940***	-5.762***	-9.857**	-2.161**	-2.411	-4.647
	(0.0370)	(0.0195)	(1.948)	(1.744)	(4.594)	(0.970)	(2.841)	(2.929)
Oversubscription				-17.11**		-18.07***		
				(8.355)		(5.366)		
Green = 0 × Oversubscription					-18.10**		-18.13***	
					(8.693)		(5.303)	
Green = 1 × Oversubscription					-14.74*		-17.93***	
					(8.215)		(5.462)	
Initial yield spread						0.911***	0.911***	0.905***
						(0.106)	(0.105)	(0.107)
Medium oversubscription bucket								-7.379
								(5.671)
High oversubscription bucket								-16.86***
								(5.556)
Green × Medium oversubscription bucket								1.375
								(3.778)
Green × High oversubscription bucket								2.706
								(3.271)
Bond Group FE		✓	✓	✓	✓	✓	✓	✓
Month FE		✓	✓	✓	✓	✓	✓	✓
Rating FE		✓	✓	✓	✓	✓	✓	✓
Bond Option FE		✓	✓	✓	✓	✓	✓	✓
Currency FE		✓	✓	✓	✓	✓	✓	✓
Bond & Market Controls		✓	✓	✓	✓	✓	✓	✓
Observations	266	260	260	260	260	260	260	260
R^2	0.014	0.948	0.977	0.978	0.978	0.992	0.992	0.992
Adjusted R^2	0.010	0.842	0.931	0.933	0.933	0.976	0.975	0.973

Regressions of corporate yield spreads at issuance on the green bond indicator variable and bond oversubscription variables, using the three-to-one matched bond sample that spans the period 2014 through 2021 (Subsection 4.2 details the matching methodology). In Columns (1) and (2), the dependent variable, *Oversubscription*, represents bonds' ratio of the notional amount of orders to the actual amount of issued debt. In Columns (3) to (8), the dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variables are defined as follows: *Green* takes the value of one for green bonds and zero otherwise. *Oversubscription* is the ratio of the notional amount of orders to the actual amount of issued debt. *Initial yield spread* represents bonds' yield spread that underwriters propose to investors at the beginning of the book-building process (as opposed to bonds' yield spread once the bond is priced on issuance day). *Low/medium/high oversubscription bucket* represent terciles formed on bonds' oversubscription ratios. All other control variables and fixed effects are detailed in Section 3 and Subsection 4.2. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

4.4.2. Index inclusion

Another potential proxy for excess demand relative to a green bond's fundamentals is whether the green bond is included in a green bond index. Index inclusion is either known in advance or can be anticipated based on inclusion criteria and the inclusion of prior issues. Previous literature shows that demand from mutual funds and ETFs benchmarked to a given bond index can impact the pricing of bonds included in the index (Ye, 2019; Dannhauser, 2017; Holden and Nam, 2019). This phenomenon of additional demand for index constituents has been shown to encourage bond issuers to alter the characteristics of their bonds in order to meet index inclusion criteria (Dathan and Davydenko, 2020; Calomiris et al., 2022). Moreover, benchmarking has even been shown to impact capital flows and exchange rates across countries, particularly emerging market economies (Raddatz et al., 2017; Pandolfi and Williams, 2019). Index inclusion may plausibly drive the greenium by forcing demand for constituent green bonds from bond funds benchmarked against the indexes. In addition, institutional investors with buy-and-hold strategies prefer to purchase bonds with a primary market allocation to avoid the substantial transaction costs associated with buying the same position in the secondary bond market (Nikolova et al., 2020; Flanagan et al., 2021). As green bonds are anecdotally reported to be less liquid than conventional bonds in secondary-market trading, this rationale should exacerbate the competition for green primary market allocations in the environmental, social, and governance (ESG) asset management industry.[43]

We hypothesize that green bond index inclusion is negatively related to yield spreads at issuance. In Table 8 Column (1), we confirm that green bonds capture a significant average greenium of about 11 basis points when they are included in any of the three green bond indexes that we consider (ICE, Solactive, JP Morgan).[44] In Column (2), we control for the green bond indicator to test if the greenium for index constituents is distinct from the advantage for excluded green bonds. We find that index inclusion is not, on average, associated with a statistically significant greenium relative to excluded bonds. However, in Columns (3) and (4), we segment the index inclusion effect by currency. We find that euro green bonds included in the green bond indexes receive a large, statistically significant greenium of about 12 basis points, while excluded euro green bonds do not. On the other hand, included U.S. dollar green bonds do not receive a greenium, while excluded dollar green bonds receive a sizable, statistically significant greenium.

[43] For a brief discussion of how (expected) secondary-market liquidity may affect the demand of institutional investors in the primary green bond market, see Section A2 in the Appendix.

[44] We do not report the results from regressions using each green bond index separately, but the results are similar.

Table 8
Greenium and bond index inclusion.

	Yield spread			
	(1)	(2)	(3)	(4)
Green × Triple index inclusion	-10.61***	-4.991		
	(2.828)	(4.710)		
Green		-5.747		
		(3.840)		
Green × EUR × Triple index inclusion			-11.79***	-12.56**
			(3.366)	(4.926)
Green × USD × Triple index inclusion			-9.158*	11.88
			(4.946)	(10.95)
Green × EUR				0.753
				(3.112)
Green × USD				-21.56**
				(10.67)
Currency × Month FE	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓
Seniority FE	✓	✓	✓	✓
Bond Option FEs	✓	✓	✓	✓
Industry & SOE FEs	✓	✓	✓	✓
Bond & Market Controls	✓	✓	✓	✓
Firm × Year FE	✓	✓	✓	✓
Observations	114,879	114,879	114,879	114,879
R^2	0.843	0.843	0.843	0.843
Adjusted R^2	0.810	0.810	0.810	0.810

Regressions of corporate yield spreads at issuance on the green bond and the bond index inclusion indicator variables, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variables are defined as follows: *Green* takes the value of one for green bonds and zero otherwise. *Triple index inclusion* is equal to one if a green bond was a constituent of three major green bond indices (ICE, Solactive, JP Morgan). *EUR* and *USD* are indicator variables that differentiate euro- and U.S. dollar-denominated green bonds. All other control variables and fixed effects are detailed in Section 3. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Notably, most of the excluded U.S. dollar green bonds have been issued by foreign dollar issuers, a green bond feature we will study in more detail in the following subsection.

In combination, the results in Tables 7a, 7b, and 8 suggest that oversubscription and bond index inclusion are relevant drivers in the formation of the greenium. Given that firms' supply of green bonds is rather inelastic, our findings suggest that firms respond to high demand for their green bonds at issuance by raising the price of their bonds (that is, lowering the yield). Our results point to a more pronounced greenium through demand pressures in euro-denominated green bonds and U.S. dollar green bonds of foreign issuers. One possible explanation for the latter effect could be the more stringent transparency requirements that European issuers and financial service providers are subjected to. This may create additional demand for the respective green bonds from ESG investors.

4.5. Greenium by currency denomination

Next, we examine the effect of currency denomination and issuer geography on the greenium. Table 9 Column (1) shows that euro-denominated green corporate bonds capture an average greenium of about 6 basis points. In comparison, U.S. dollar-denominated green bonds capture an average greenium of 13 basis points. In Column (2), we show that the roughly 6 basis point difference in the greenium for euro and U.S. dollar green bonds is not statistically significant from zero, meaning the bonds capture statistically comparable greeniums when the point estimates' confidence intervals are taken into account.[45]

We further segment the green bond sample by currency and whether a bond was issued by a domestic or foreign firm relative to the currency region. In Column (3), our findings suggest that, on average, the greenium for U.S. dollar green bonds is conferred to foreign issuers, not American issuers. This is consistent with Table 8 Column (3), as most of the dollar green bonds excluded from green bond indexes in our sample are issued by foreign firms. In Column (4), we show that the 15 basis point difference between local and foreign issuers in the U.S. dollar green bond market is significant at the 5% significance level. On the other hand, the euro greenium appears to be conferred primarily to local issuers (Column (3)). However, compared to the U.S. dollar green bond market, we cannot confirm a statistically significant difference in the greenium of local and foreign euro green bond issuers (Column (4)).

In Column (5), we examine the impact of issuer geography on the greenium. We split green bond issuers into five groups: the euro area, the United States, other developed markets, China, and other emerging markets.[46,47] We find significant statistical evidence for a 6 basis point average greenium for euro area issuers and a 21 basis point average greenium for other developed market issuers, but no statistical evidence of a greenium for issuers from the United States. As shown in Column

[45] We find quantitatively and qualitatively similar estimates for all results in Table 9 when we run regressions on a subsample that excludes all but U.S. dollar- and euro-denominated bonds.

[46] The issuer country is the country of the issuer's ultimate parent as of 2021.

[47] Results remain qualitatively similar when we use the EU instead of the euro area.

Table 9
Greenium and currency denomination and region.

	Yield spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Green × EUR	-5.696***			-4.295		
	(2.136)			(5.294)		
Green × USD	-13.46***	-7.768		-15.64**		
	(5.110)	(5.507)		(6.228)		
Green		-5.696***				-6.209**
		(2.136)				(2.388)
Green × Local EUR			-6.019***	-1.725		
			(2.194)	(5.372)		
Green × Foreign EUR			-4.295			
			(5.294)			
Green × Local USD			-6.579	9.065		
			(10.02)	(12.22)		
Green × Foreign USD			-15.64**			
			(6.228)			
Green × China					-7.590	-1.381
					(8.166)	(8.151)
Green × Euro area					-6.209**	
					(2.388)	
Green × USA					-2.652	3.557
					(7.819)	(8.315)
Green × Other DM					-21.06**	-14.85
					(9.118)	(9.318)
Green × Other EM					-15.35	-9.143
					(16.25)	(16.75)
Currency × Month FE	✓	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Seniority FE	✓	✓	✓	✓	✓	✓
Bond Option FEs	✓	✓	✓	✓	✓	✓
Industry & SOE FEs	✓	✓	✓	✓	✓	✓
Bond & Market Controls	✓	✓	✓	✓	✓	✓
Firm × Year FE	✓	✓	✓	✓	✓	✓
Observations	114,879	114,879	114,879	114,879	114,879	114,879
R^2	0.843	0.843	0.843	0.843	0.843	0.843
Adjusted R^2	0.810	0.810	0.810	0.810	0.810	0.810

Regressions of corporate yield spreads at issuance on the green bond and currency denomination indicator variables, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variables are defined as follows: *Green* takes the value of one for green bonds and zero otherwise. *EUR* and *USD* are indicator variables that differentiate euro- and U.S. dollar-denominated green bonds. *Local* and *Foreign* differentiate currency denomination by local and foreign currency issuers. *China*, *Euro area*, *USA*, *Other DM*, and *Other EM* are indicator variables reflecting issuer geography. All other control variables and fixed effects are detailed in Section 3. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

(6), we cannot confirm that these differences in the greenium across regions are statistically significant. Overall, the results from Table 9 suggest that, on average, euro area firms issuing local currency bonds as well as foreign-currency dollar borrowers receive a significant average greenium, while there is no average greenium in the domestic dollar market and in the non-euro area market more generally.

4.6. Green bond governance, external review, and credibility

In Table 10, we analyze whether the "greenness" of a green bond has a measurable impact on its yield. We begin by assessing the impact of a green bond's governance and alignment with voluntary international standards. In Column (1), we separate the green bond sample into those that are aligned and not aligned with the GBP. In our sample, 82% (965 of 1,169) of green bonds are aligned with the GBP. Should the marginal green bond investors prefer GBP alignment, one would expect that unaligned bonds are less attractive and therefore receive a lower greenium. We find that, on average, GBP-aligned green bonds capture a robust statistically significant greenium of 8 basis points, while non-GBP-aligned green bonds show a coefficient of about 22 basis points that is weakly significant at the 10% significance level. In Column (2), we check whether there is a statistically significant or measurable difference between the two groups of bonds, which we cannot confirm. The lack of a statistically significant difference between the two groups suggests that their pricing is not inherently different and that green bonds with poor governance frameworks may still receive a comparable borrowing cost advantage.

Next, we separate green bonds into those that do and do not receive an external review (also known as third-party verification). In our sample, 78% (913 of 1,169) of green bonds have received an external review. Like above, our hypothesis is that green bonds without an external review should receive a lower greenium if the marginal green bond investor cares about third party reviews. In Column (3), we find that, on average, green bonds with an external review capture a statistically significant greenium of 8 basis points, while green bonds without an external review do not capture a significant greenium despite a point

Table 10
Greenium and bond governance, external review, and proceed credibility.

	Yield spread					
	(1)	(2)	(3)	(4)	(5)	(6)
Green × GBP aligned	-7.522***					
	(2.314)					
Green × GBP not aligned	-21.63*	-14.11				
	(11.57)	(11.41)				
Green		-7.522***		-8.454***		-6.651**
		(2.314)		(2.553)		(2.775)
Green × External review			-8.454***			
			(2.553)			
Green × No external review			-9.898	-1.444		
			(8.196)	(8.390)		
Green × No refinancing					-6.651**	
					(2.775)	
Green × Refinancing					-13.35**	-6.699
					(5.442)	(5.475)
Currency × Month FE	✓	✓	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓	✓	✓
Seniority FE	✓	✓	✓	✓	✓	✓
Bond Option FEs	✓	✓	✓	✓	✓	✓
Industry & SOE FEs	✓	✓	✓	✓	✓	✓
Bond & Market Controls	✓	✓	✓	✓	✓	✓
Firm × Year FE	✓	✓	✓	✓	✓	✓
Observations	114,879	114,879	114,879	114,879	114,403	114,403
R^2	0.843	0.843	0.843	0.843	0.843	0.843
Adjusted R^2	0.810	0.810	0.810	0.810	0.809	0.809

Regressions of corporate yield spreads at issuance on the green bond indicator variable and variables capturing green bond governance, external review, and credibility, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variables are defined as follows: *Green* takes the value of one for green bonds and zero otherwise. *GBP aligned* is equal to zero if a green bond is not fully aligned with the Green Bond Principles and one if it is fully aligned. *External review* equals one if a green bond was subject to a pre-issuance external review by a third party and zero otherwise. *No refinancing* equals one if some portion of the bond was used to refinance an existing liability and zero otherwise. All other control variables and fixed effects are detailed in Section 3. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

estimate of about 9 basis points. Again, as shown in Column (4), we do not find a statistically significant difference in the greenium of green bonds with and without an external review. This suggests that green bonds without an external review may still capture a borrowing cost advantage, even though it is, on average, not significantly different from zero.

Lastly, in Columns (5) and (6), we separate green bonds by whether the bond has been used to refinance a firm's existing debt. For green bonds that are not used for refinancing, we estimate a greenium of about 7 basis points that is significant at the 5% significance level. For green bonds used for refinancing, the greenium is about 13 basis points and also significant at the 5% significance level. We show in Column (6) that this 7 basis point difference between refinancing and non-refinancing green bonds is not statistically significant, indicating no measurable impact of the refinancing variable on the greenium. This result is notable because it suggests that whether a green bond drives additional environmental impact from new green projects, as opposed to refinancing existing projects, is not priced in the primary market.[48]

In combination, the results from Table 10 suggest that the greenium is primarily allocated to green bonds that are sufficiently green beyond some minimum threshold. That is, on the margin, it appears that what determines the greenium is not the credibility of the green promise, but rather the requirements that make bonds "green enough" (full or partial GBP alignment and external review) to be included in a conventional ESG or green bond portfolio Heeb et al. (2022).

4.7. Bond- and firm-level characteristics

This last subsection analyzes the connection between bond- and firm-level characteristics and the greenium. Table 11 presents the effect of bond size, firm capital market access, and bond rating on the greenium. In Column (1), we do not find that the size of the green bond itself impacts the greenium. However, in Column (2), we estimate the greenium while controlling for the average size of an issuer's bonds within the same year. We find that, on average, firms that can place larger bonds—a proxy for access to debt capital markets—capture a larger greenium. Specifically, the coefficient on the green bond indicator decreases in the average size of an issuer's bonds. For the sample mean of log average bond size, 18.8 (about $152 million), the greenium is about 7 basis points (58.4 minus 3.5 times 18.8).

In Column (3), we assess the impact of a green bond's rating on the greenium, segmenting the sample into investment-grade (AAA to BBB), high-yield (BB to C), and unrated bonds. Only investment-grade green bonds capture a statistically significant average greenium of about 10 basis points. In contrast, the greenium coefficients for high-yield and unrated green bonds are statistically insignificant. In Column (4), we

[48] Green investments are said to have an additional environmental impact (in ESG jargon, "additionality") when they produce beneficial outcomes that would not have occurred without the investment. For a review of environmental and social impact investing, see Brest and Born (2013).

Table 11
Greenium and bond- and firm-level characteristics.

	Yield spread			
	(1)	(2)	(3)	(4)
Green	20.68	57.71**		-10.34***
	(26.36)	(28.59)		(3.035)
Green × Size	-1.531			
	(1.396)			
Green × Average issuer bond size		-3.458**		
		(1.509)		
Green × Investment-grade			-10.34***	
			(3.035)	
Green × High-yield			-25.33	-14.99
			(18.58)	(19.09)
Green × Not rated			-1.571	8.767*
			(3.258)	(4.897)
Currency × Month FE	✓	✓	✓	✓
Rating FE	✓	✓	✓	✓
Seniority FE	✓	✓	✓	✓
Bond Option FEs	✓	✓	✓	✓
Industry & SOE FEs	✓	✓	✓	✓
Bond & Market Controls	✓	✓	✓	✓
Firm × Year FE	✓	✓	✓	✓
Observations	114,879	114,879	114,879	114,879
R^2	0.843	0.843	0.843	0.843
Adjusted R^2	0.810	0.810	0.810	0.810

Regressions of corporate yield spreads at issuance on the green bond indicator variable and variables capturing bond and issuer characteristics, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variables are defined as follows: *Green* takes the value of one for green bonds and zero otherwise. *Size* is the logarithm of the bond's notional amount issued. *Average issuer bond size* is the average size of an issuer's bonds issued within the same year of the green bond. *Investment-grade*, *High-yield*, and *No rated* are indicator variables that flag the respective bond rating tiers. All other control variables and fixed effects are detailed in Section 3. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

do not find significant differences in the greenium between rating tiers despite the differences in group means. The difference of 8 basis points between investment-grade and unrated bonds is weakly significant, suggesting that, on average, investment-grade bonds have a larger greenium than unrated bonds. This finding indicates that firms that cannot afford to have their bonds rated or firms not covered by rating agencies (for instance, SMEs and emerging market issuers) may not capture a greenium. These results are consistent with our findings in Table 8, as they point to a greenium for firms that are able to meet typical index inclusion criteria – namely, issuers with an ability to issue large, investment-grade rated bonds regularly.

In Table 12, we analyze how the greenium is allocated by issuer sector. In Column (1), we find that primarily issuers from the banking sector receive a statistically significant average greenium of about 8 basis points. A possible explanation for this banking greenium could be that banks receive compensation for certifying, extending, and monitoring green loans to their customers or that they capture a potential borrowing cost advantage that may be passed through to the banks' green debtors. What's more, to prepare for anticipated climate-related financial disclosures, the banking sector has increased its climate disclosure efforts, which may increase the attractiveness of banks' bonds to investors who are subject to sustainability-related reporting requirements. Most other sectors receive weakly significant (electric utilities and fossil fuels, transportation) or statistically insignificant negative coefficient estimates. The notable exception is the high-emission industry and materials sector, for which we estimate a positive yet statistically insignificant coefficient.

In Column (2), we do not find statistically significant differences in the greenium between banks and issuers in other sectors, suggesting that other sectors show statistically comparable greeniums when the estimates' confidence intervals are considered. The notable exception is the difference in the greenium between banks and industry and materials. This difference is about 23 basis points and is weakly significant at the 10% significance level.

In Table A4 in the Appendix, we further break down the greenium in the banking sector by region. We find that it is primarily banks from emerging markets – in particular, China – as well as banks from the EU that capture a significant greenium.[49] The significant greenium estimate for EU banks aligns with our overall results on a robust euro greenium for large, investment-grade issuers. It is also consistent with the idea that increased climate disclosures, particularly in Europe, increase the attractiveness of the region's bonds relative to other developed markets. For emerging markets, where information asymmetries can be significant and access to issuers can be limited, the explanation that the estimated greenium could be compensating banks for their monitoring costs and access to green debtors applies as well. Lastly, nearly all of the emerging markets and Chinese banks' green bonds have full GBP alignment and an external review (Deng et al., 2020; Tang et al., 2023; Li et al., 2022; Hu et al., 2022) as well as a high credit rating, which can make the bonds more likely for inclusion in international bond indexes

[49] Besides the bond rating, our regression specification controls for whether an issuer is an SOE through the issuer fixed effect. It also accounts for differences in state ownership across jurisdictions with a country-times-SOE interaction. In unreported results, this effect remains qualitatively similar when we drop all SOEs from the sample.

Table 12
Greenium and issuer industry.

	Yield spread	
	(1)	(2)
Green		-8.494***
		(2.910)
Green × Alternate energy	-31.82	-23.32
	(26.29)	(26.08)
Green × Banks	-8.494***	
	(2.910)	
Green × Electric utilities and fossil fuels	-15.43*	-6.941
	(7.932)	(8.834)
Green × Industry and materials	14.54	23.03*
	(11.33)	(11.84)
Green × Non-bank financials	-3.109	5.385
	(14.03)	(14.08)
Green × Real estate	-1.601	6.893
	(9.228)	(9.791)
Green × Transportation	-13.03	-4.533
	(8.678)	(9.600)
Green × Other	-8.718*	-0.224
	(5.014)	(5.633)
Currency × Month FE	✓	✓
Rating FE	✓	✓
Seniority FE	✓	✓
Bond Option FEs	✓	✓
Industry & SOE FEs	✓	✓
Bond & Market Controls	✓	✓
Firm × Year FE	✓	✓
Observations	114,879	114,879
R^2	0.843	0.843
Adjusted R^2	0.810	0.810

Regressions of corporate yield spreads at issuance on the green bond indicator and issuer industry indicator variables, using a sample that spans the period 2014 through 2021 (Section 3 details the sample construction). The dependent variable, *Yield spread*, refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variables are defined as follows: *Green* takes the value of one for green bonds and zero otherwise. *Alternative energy*, *Banks*, *Electric utilities and fossil fuels*, *Industry and materials*, *Non-bank financials*, *Real estate*, *Transportation*, and *Other* are indicator variables that flag the respective issuer industries. All other control variables and fixed effects are detailed in Section 3. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

and, thus, attractive for portfolio diversification and index tracking for a sustainable asset manager.

5. Conclusion

Green corporate bonds have emerged as one of the most recognized financing instruments for firms looking to invest in environmentally sustainable projects. This paper's findings hold pertinent implications for the role of corporate green bonds in incentivizing meaningful green investments throughout the global economy. While our empirical evidence suggests that a small but significant borrowing cost advantage over conventional bonds exists, it primarily favors large, rated, local euro and foreign U.S. dollar issuers from developed economies, does not necessarily reward high-quality green projects, and is likely small or potentially negative when taken net of fees and compliance costs. What's more, part of the funding advantage of green corporate bonds may be driven by potentially temporary supply and demand imbalances between green bond issuers and the growing sustainable asset management industry.[50] Furthermore, questions of additionality – whether green bonds incentivize additional green investment or merely reward green investments that would have occurred anyway – remain unresolved.

For these reasons, the borrowing cost advantage of green corporate bonds likely plays a limited role in incentivizing large-scale climate investment. Instead, the primary benefit of issuing a green bond is likely an indirect signaling benefit that improves an issuer's environmental credentials. This raises questions about the quality of the signal, though. While issuing a green bond may indicate that a firm is greener than its peers, there is a risk of greenwashing and no guarantee that funded green projects reflect a sustained commitment to meaningful green investment.

Mandated environmental reporting and disclosures of climate-related risks may be a more effective signaling solution (Bolton and Kacperczyk, 2021). Financial instruments linked to firm-level environmental metrics, such as sustainability-linked bonds rather than specific green projects (like green bonds), may be another option for connecting sustainability-minded investors with firms making credible, systemic green investments. Situated within the need for substantial, rapid investments to reach net zero emissions by 2050, an overreliance on the potentially limited incentives conferred from issuing green bonds could lead to an undersupply of green investment if green bonds are not complemented by other sources of private-sector funding.

CRediT authorship contribution statement

John Caramichael: Formal analysis, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing, Data curation, Visualization. **Andreas C. Rapp:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Supervision, Writing – original draft, Writing – review & editing.

Data availability

The authors do not have permission to share data.

Appendix A

A1. Motivations for investing in green bonds

While our paper takes the firm's perspective, this section briefly outlines the investor's perspective by discussing several motives for investing in green bonds.

There are several explanations for why one would accept a lower yield on a green bond relative to a comparable conventional bond. The most common explanation for the greenium is that investors are willing to sacrifice immediate financial returns in exchange for an environmental benefit. According to this framework, because a green bond is effectively comprised of a conventional bond and a green promise, investors assign a positive value to the promise and are willing to pay a higher price for a bond at issuance, which means they accept a lower yield. Under this working model, the marginal green bond investor is considered concessionary (as opposed to full return), meaning they are willing to accept lower investment returns to finance high-quality green projects.[51] By optimizing the trade-off between returns and environmental impact, these investors are expected to exert a demand pressure

[50] Continued growth in green corporate bond issuance could, thus, lead to a dilution of the greenium, should supply catch up with demand. This potential dilution has been noted anecdotally by market participants and financial journalists (see, for example, Bahceli (2021); Stubbington (2021)).

[51] A recent draft methodology released by the Partnership for Carbon Accounting proposes that green bond investors should be able to claim a reduction to their Scope 3 Category 15 (investment-related) greenhouse gas emissions

Table A1
Bond sample, by currency.

	Conventional		*Green*	
Currency	Amt. Issued ($B)	*N*	Amt. Issued ($B)	*N*
AUD	243.0	2,109		
BRL	13.5	361		
CAD	605.4	2,386		
CHF	397.3	1,763		
CNY	4,161.0	21,572		
DKK	24.7	313		
EUR	6,708.6	22,094	302.8	663
GBP	542.1	1,222		
HKD	67.7	1,226		
IDR	54.3	1,572		
ILS	40.7	304		
INR	515.5	9,146		
JPY	897.6	6,583		
KRW	824.3	14,440		
MXN	25.8	191		
MYR	126.4	2,191		
NOK	69.6	874		
NZD	37.0	476		
RUB	61.7	660		
SEK	311.6	790		
SGD	57.5	451		
USD	17,768.9	39,239	187.4	506
ZAR	9.3	249		
Total	33,563.6	130,212	490.3	1,169

Breakdown of the bond sample by currency. The table gives the currency breakdown for conventional bonds *(left)* and green bonds *(right)*. *Amt. Issued ($B)* represents the total issuance volume of green bonds and conventional bonds, respectively, by currency, reported in U.S. dollar billions. *N* gives the number of conventional bonds and green bonds, respectively, by currency. To insure sufficient issuance volumes for a meaningful comparison, we only include conventional bonds denominated in currencies that have at least $10 billion in total issuance in the respective currency over the sample period. We also require green bonds to be issued in either the euro or the U.S. dollar because these green bond markets are the deepest and most liquid. The bond sample construction is detailed in Section 2.

that produces a "smart greenium," rewarding credible, high-impact green bonds with an incrementally lower interest rate.

Besides the smart greenium explanation, there are two main alternative explanations for the greenium. The first is that green bonds are more valuable because they are expected to achieve higher risk-adjusted returns than conventional bonds, as they are issued by firms that reduce their exposure to climate- and other environment-related physical and transition risks. This explanation is problematic because the coupon and principal payments from green bonds are rarely paid out from cash flows generated by the green bond's underlying green projects. Instead, the credit risk of a green bond should be considered identical to the credit risk of a firm's conventional bonds, particularly when the bonds have the same recourse. Even if a firm's issuance of a green bond reduces its overall credit risk by lowering its exposure to sustainability risk factors, this would not necessarily improve the risk-adjusted returns expected for green bonds, especially compared to conventional bonds.

The second explanation for the greenium, called the "crisis premium" theory, is that green bonds offer better returns during crisis periods because environmentally conscious investors are less likely to sell green bonds during market distress. For this explanation to hold, it would have to be true that green investors are, at the margin, less risk-averse

in accordance with their investment in a given green bond. This accounting methodology could potentially increase the incentive to pay a higher price for a given green bond (concede lower expected returns) by effectively allowing green bond investors to purchase an emissions reduction by buying a green bond (Partnership for Carbon Accounting Financials, 2021). Some market participants reportedly already use this methodology (Edwards et al., 2022).

Table A2
Green bond sample, by sector.

Sector	Amt. issued ($B)	*N*
Alternate energy	19.7	170
Banks	162.4	450
Electric utilities and fossil fuels	121.3	188
Industry and materials	31.1	75
Non-bank financials	24.4	47
Real estate	74.9	156
Transportation	20.6	29
Other	35.8	54
Total	490.3	1,169

Breakdown of the *green* bond sample by industry. Sector categories are based on the Global Industry Classical Standard (GICS) codes, tailored to the major segments of the green bond market. *Amt. Issued ($B)* represents the total issuance volume of green bonds by industry, reported in U.S. dollar billions. *N* gives the number of green bonds by industry. The variable construction is detailed in Section 3.

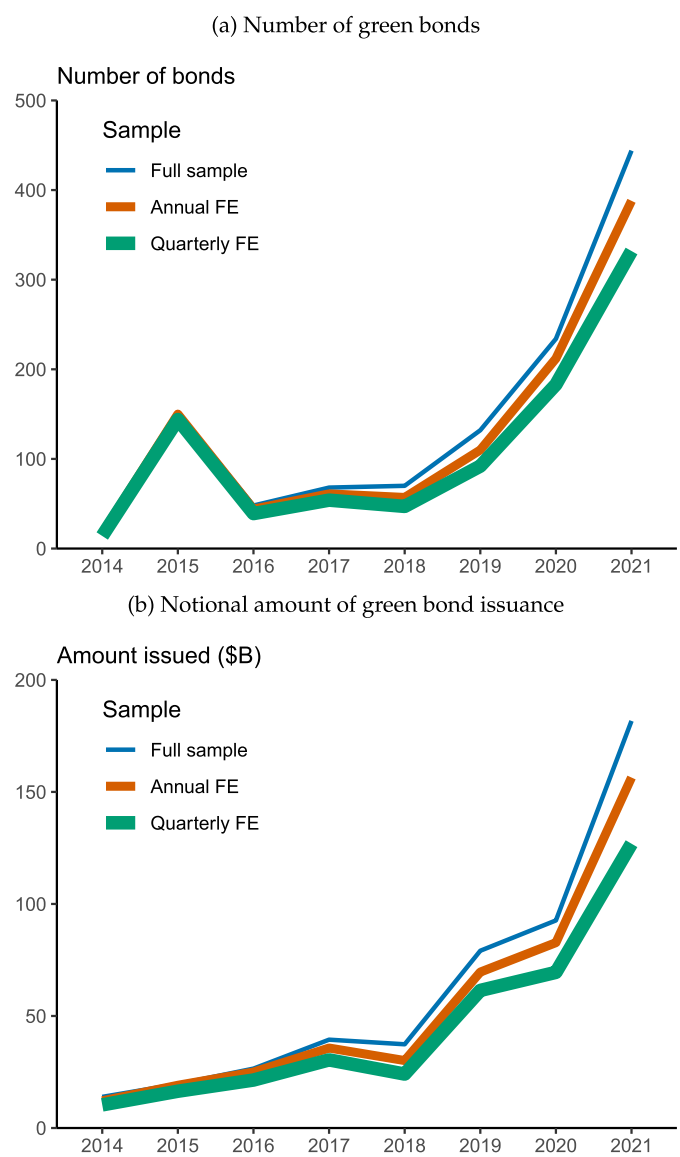

Fig. A1. Impact of time-varying fixed effects on green bond sample size. Panel (a) is the number of green bonds in our sample. Panel (b) is the notional amount of green bond issuance in our sample. The data are reported at an annual frequency. *Full sample* is the sample when we do not include time-varying fixed effects. *Annual FE* and *Quarterly FE* are the samples when we include annual or quarterly time-varying fixed effects, respectively.

Table A3
Greenium and time variation.

	Yield spread (1)
2014 × Green	-7.323
	(11.96)
2015 × Green	-2.179
	(5.358)
2016 × Green	-10.74
	(8.470)
2017 × Green	-6.163
	(8.318)
2018 × Green	3.751
	(5.248)
2019 × Green	-15.33**
	(6.373)
2020 × Green	-10.02***
	(3.544)
2021 × Green	-8.799**
	(3.784)
Currency × Month FE	✓
Rating FE	✓
Seniority FE	✓
Bond Option FEs	✓
Industry & SOE FEs	✓
Bond & Market Controls	✓
Firm × Year FE	✓
Observations	114,879
R^2	0.843
Adjusted R^2	0.810

Regressions of corporate yield spreads at issuance on the *Green* bond indicator variable interacted with annual indicator variables for the years 2014 to 2021 (Section 3 details the sample construction). The dependent variable *Yield spread* refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variable *Green* takes the value of one for green bonds and zero otherwise. *2014* to *2021* are indicator variables that flag the respective calendar year. All other control variables and fixed effects are detailed in Section 3. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Table A4
Greenium and issuer industry, banks by region.

	Yield spread	
	(1)	(2)
Green		-23.82***
		(5.953)
Green × Banks × China	-23.82***	
	(5.957)	
Green × Banks × EU	-6.178***	17.64***
	(2.326)	(6.167)
Green × Banks × USA	12.16	35.98***
	(8.389)	(11.17)
Green × Banks × Other DM	-1.016	22.80**
	(8.784)	(11.27)
Green × Banks × Other EM	-71.77*	-47.95
	(38.00)	(38.25)
Green × Non-bank financials	-3.279	20.54
	(14.04)	(15.29)
Green × Non-financials	-9.514*	14.31**
	(5.050)	(6.978)
Currency × Month FE	✓	✓
Rating FE	✓	✓
Seniority FE	✓	✓
Bond Option FEs	✓	✓
Industry & SOE FEs	✓	✓
Bond & Market Controls	✓	✓
Firm × Year FE	✓	✓
Observations	114,879	114,879
R^2	0.843	0.843
Adjusted R^2	0.810	0.810

Regressions of corporate yield spreads at issuance on the *Green* bond indicator and issuer industry and region variables for the years 2014 to 2021 (Section 3 details the sample construction). The dependent variable *Yield spread* refers to a bond's yield spread on its date of issuance over the maturity-matched government bond yield for the given bond's currency region. The independent variable *Green* takes the value of one for green bonds and zero otherwise. *Banks*, *Non-bank financials*, and *Non-financials* are indicator variables that flag the respective issuer industries. *China*, *EU*, *USA*, *Other DM*, and *EM* are indicator variables reflecting issuer geography. All other control variables and fixed effects are detailed in Section 3. Standard errors, reported in parentheses, are clustered on the issuer ultimate parent and year-month levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

than conventional investors or have considerably longer investment horizons to sit out temporary dislocation in bond prices. This seems plausible for institutional investors, who may trade off selling sustainable assets against the perceived penalty to their cost of capital from failing to hit their sustainable investment goals, typically articulated in an ESG shareholder engagement. However, this seems less plausible for retail investors, who are typically more risk-averse and may view green investments as a luxury preference that withers under financial or economic stress.[52] While the crisis premium theory should be considered, the limited available market participant commentary suggests this theory is not widely held or inconclusive.[53]

A2. The greenium and (expected) secondary market liquidity

In this section, we briefly outline how (expected) secondary market liquidity may affect institutional investors' demand in the primary green bond market. Our rationale builds on empirical evidence by Nikolova et al. (2020) and Flanagan et al. (2021) linking greater competition for primary market allocations among buy-and-hold investors to secondary market illiquidity.

A2.1. Expected secondary market liquidity

Assume the yield for a given bond can be decomposed into three components such that

$$y = y_F + y_D + y_{EL} \tag{A1}$$

where y_F is the compensation for a bond's fundamental characteristics; y_D is the compensation for investor demand for a particular bond type beyond its fundamentals; and y_{EL} is a compensation for a bond's expected liquidity in the secondary market. For now, assume that y_D and y_{EL} are independent of each other. The bond's fundamental characteristics (for instance, credit rating, coupon, and maturity) determine its risk profile across several key dimensions, such as credit risk, interest rate risk, and inflation risk. A given bond with riskier fundamentals will have higher values of y_F (lower price p_F). Beyond the fundamentals, excess demand from institutional investors for a given bond type will exert demand pressure on the bond, lowering the value of y_D (raising price p_D). Finally, lower expected liquidity in the secondary market should raise y_{EL} (lower the price p_{EL}) because bonds that are more difficult to sell or buy in the secondary market require higher compensation as they are tied to high transaction costs.

When measuring the greenium in our regressions, we compare the yield on a green bond to that of a comparable conventional bond (observed or hypothetical) with the same fundamentals. In other words, we test whether $y_F^{Green} = y_F^{Conventional}$ for comparable bonds. Any yield

[52] For example, Döttling and Kim (2020) find evidence that the sustainability preferences of retail investors deteriorated relative to institutional investors during the market crash of March 2020.

[53] See, for example, Edwards et al. (2020) and Marsh (2020).

differential between a green bond and a comparable conventional bond should then be given by the following components:

$$\underbrace{y^{Green} - y^{Conventional}}_{\text{Yield differential}} = \underbrace{\left(y_D^{Green} - y_D^{Conventional}\right)}_{\text{Greenium (-)}} + \underbrace{\left(y_{EL}^{Green} - y_{EL}^{Conventional}\right)}_{\text{Expected liquidity differential (+)}} \tag{A2}$$

We argue that $y_D^{Green} < y_D^{Conventional}$ because excess demand for the green bond, for example, from the growing sustainable asset management industry, lowers its yield relative to the yield for the conventional bond. Therefore, the greenium term is negative. However, if a green bond has a lower expected liquidity than a comparable conventional bond because green bond investors are more likely to be buy-and-hold investors, then $y_{EL}^{Green} > y_{EL}^{Conventional}$, and the expected liquidity term will be positive. Controlling for relevant bond characteristics, we find that the green bond yield differential, given by $y^{Green} - y^{Conventional}$, is indeed negative. If there is an expected liquidity penalty for green bonds beyond their fundamentals, meaning the expected liquidity term is positive, we may be underestimating the magnitude of the "true" greenium by not controlling for differences in expected liquidity.[54] However, even if this were the case, it would not impact the ultimate borrowing cost from the issuer's perspective, measured by the yield differential $y^{Green} - y^{Conventional}$. It merely affects the attribution of the yield differential to its various components.

A2.2. Decomposing the effect of poor expected liquidity

To further clarify the effect of poor expected secondary market liquidity and outline the possible correlation between the demand and the liquidity components, let us decompose the yield further. Above, we let y_{EL} refer to investor compensation for expected secondary market liquidity, for instance, for high expected transaction costs of selling the bond before it matures. As before, lower secondary market liquidity would reduce the overall price p_{EL} of the bond (raise yield y_{EL}). Now, if investors believe it will be costly to buy the bond in the secondary market, they may, at the margin, be more willing to buy the bond in the primary market. Thus, expected illiquidity contributes to the excess demand for the bond type beyond the bond's fundamentals. Let us denote the yield compensation from this primary market excess demand due to liquidity considerations as y_{PM}, where a stronger preference for a primary market allocation lowers y_{PM} (raises y_{PM}). Let us re-label the residual demand for the bond type, uncorrelated with expected bond liquidity, as y_{RD}. Thus, the yield differential from Equation (A2) can be rewritten:

$$\underbrace{y^{Green} - y^{Conventional}}_{\text{Yield differential}} = \underbrace{\left(y_{RD}^{Green} - y_{RD}^{Conventional}\right)}_{\text{Greenium (-)}} + \underbrace{\left(y_{EL}^{Green} - y_{EL}^{Conventional}\right)}_{\text{Expected liquidity differential (+)}} + \underbrace{\left(y_{PM}^{Green} - y_{PM}^{Conventional}\right)}_{\text{Primary market demand (-)}} \tag{A3}$$

If a green bond has a lower expected liquidity in the secondary market than a comparable conventional bond, this should increase the preference to buy the green bond in the primary market. This means $y_{PM}^{Green} < y_{PM}^{Conventional}$, so the additional primary market term will still be negative. As in the previous subsection, note that while this may affect the attribution of the greenium in Equation (A3) to the various components, but it ultimately does not impact the borrowing cost of the issuer to the extent that the additional primary market demand is driven by the green label itself.

[54] Data availability limits our ability to construct measures of expected liquidity to empirically control for it at issuance across the bonds in our global dataset.

References

Alfaro, L., Asis, G., Chari, A., Panizza, U., 2019. Corporate debt, firm size and financial fragility in emerging markets. J. Int. Econ. 118, 1–19.

Austin, P.C., 2009. Some methods of propensity-score matching had superior performance to others: results of an empirical investigation and Monte Carlo simulations. Biom. J., J. Math. Methods Biosci. 51 (1), 171–184.

Avdjiev, S., Chui, M.K., Shin, H.S., 2014. Non-financial corporations from emerging market economies and capital flows. BIS Q. Rev.

Ayala, D., Nedeljkovic, M., Saborowski, C., 2017. What slice of the pie? The corporate bond market boom in emerging economies. J. Financ. Stab. 30, 16–35.

Bachelet, M.J., Becchetti, L., Manfredonia, S., 2019. The green bonds premium puzzle: the role of issuer characteristics and third-party verification. Sustainability 11 (4), 1098.

Bahceli, Y., 2021. Analysis: 'Greenium' shrinks as climate bond sales swell to record. Reuters.

Baker, M., Bergstresser, D., Serafeim, G., Wurgler, J., 2022. The pricing and ownership of US green bonds. Annu. Rev. Financ. Econ. 14, 415–437.

Baker, M., Wurgler, J., 2002. Market timing and capital structure. J. Finance 57 (1), 1–32.

Bao, J., Pan, J., Wang, J., 2011. The illiquidity of corporate bonds. J. Finance 66 (3), 911–946.

Bauer, R., Hann, D., 2010. Corporate environmental management and credit risk. Working Paper (SSRN 1660470).

Baulkaran, V., 2019. Stock market reaction to green bond issuance. J. Asset Manag. 20 (5), 331–340.

Becker, B., Ivashina, V., 2015. Reaching for yield in the bond market. J. Finance 70 (5), 1863–1902.

Beltran, D., Garud, K., Rosenblum, A., 2017. Emerging Market Nonfinancial Corporate Debt: How Concerned Should We Be? IFDP Notes. Board of Governors of the Federal Reserve System, Washington.

Bloomberg Back Office Fixed Income Corporates Reference Data, 2021. Bloomberg Finance LP.

Bloomberg Global Corporate Structure, 2021. Bloomberg Finance LP.

Bloomberg Valuation Services (BVAL), 2021. Bloomberg Finance LP.

BloombergNEF, 2021. Bloomberg Finance LP.

Bolton, P., Chen, H., Wang, N., 2013. Market timing, investment, and risk management. J. Financ. Econ. 109 (1), 40–62.

Bolton, P., Kacperczyk, M.T., 2021. Carbon disclosure and the cost of capital. Working Paper (SSRN 3755613).

Brest, P., Born, K., 2013. Unpacking the impact in impact investing. Stanf. Soc. Innov. Rev. 11.

Bruno, V., Shin, H.S., 2020. Currency depreciation and emerging market corporate distress. Manag. Sci. 66 (5), 1935–1961.

Calomiris, C.W., Larrain, M., Schmukler, S.L., Williams, T., 2022. Large international corporate bonds: investor behavior and firm responses. J. Int. Econ. 137, 103624.

Choi, J., Kronlund, M., 2018. Reaching for yield in corporate bond mutual funds. Rev. Financ. Stud. 31 (5), 1930–1965.

Climate Bonds Initiative, 2021. Green Bond Pricing in the Primary Market: January - June 2021.

Correia, S., 2015. Singletons, cluster-robust standard errors and fixed effects: a bad mix. Technical Note. Duke University.

Dannhauser, C.D., 2017. The impact of innovation: evidence from corporate bond exchange-traded funds (ETFs). J. Financ. Econ. 125 (3), 537–560.

Dathan, M., Davydenko, S., 2020. Debt issuance in the era of passive investment. Working Paper (SSRN 3152612).

Daubanes, J.X., Mitali, S.F., Rochet, J.-C., 2021. Why do firms issue green bonds? In: Swiss Finance Institute Research Paper, 21.

Deng, Z., Tang, D.Y., Zhang, Y., 2020. Is "greenness" priced in the market? Evidence from green bond issuance in China. J. Altern. Invest. 23 (1), 57–70.

Döttling, R., Kim, S., 2020. Sustainability preferences under stress: mutual fund flows during Covid-19. VoxEU.org 19.

Edwards, C., Harju, J., Aksu, C., Foux, M., Herold, F., 2020. Green Is but a Colour. Barclays.

Edwards, C., Rekrut, D., Boulanger, C., Elias, K., O'Neal, M., Boyipadu, P., Pobjoy, J., 2022. A Compilation of ESG Bond Trade Ideas. Barclays.

Ehlers, T., Mojon, B., Packer, F., 2020. Green bonds and carbon emissions: exploring the case for a rating system at the firm level. BIS Q. Rev.

Ehlers, T., Packer, F., 2017. Green bond finance and certification. BIS Q. Rev.

Fancy, T., 2021. Tariq Fancy on the failure of green investing and the need for state action. Economist.

Flammer, C., 2021. Corporate green bonds. J. Financ. Econ. 142.

Flanagan, T., Kedia, S., Zhou, X.A., 2021. Assessing gains from primary market allocations in corporate bonds. Working Paper (SSRN 3859063).

Ge, W., Liu, M., 2015. Corporate social responsibility and the cost of corporate bonds. J. Account. Public Policy 34 (6), 597–624.
Gianfrate, G., Peri, M., 2019. The green advantage: exploring the convenience of issuing green bonds. J. Clean. Prod. 219, 127–135.
Gilchrist, S., Zakrajšek, E., 2012. Credit spreads and business cycle fluctuations. Am. Econ. Rev. 102.
Hachenberg, B., Schiereck, D., 2018. Are green bonds priced differently from conventional bonds? J. Asset Manag. 19 (6), 371–383.
Hameed, A., Helwege, J., Li, R., Packer, F., 2019. Measuring corporate bond liquidity in emerging market economies: price- vs quantity-based measures. BIS Paper 102e.
Heeb, F., Kölbel, J.F., Paetzold, F., Zeisberger, S., 2022. Do investors care about impact?. Rev. Financ. Stud. 31 (5), 1737–1787.
Holden, C.W., Nam, J., 2019. Market accessibility, corporate bond ETFs, and liquidity. In: Kelley School of Business Research Paper, 3083257.
Houweling, P., Mentink, A., Vorst, T., 2005. Comparing possible proxies of corporate bond liquidity. J. Bank. Finance 29 (6), 1331–1358.
Hu, X., Zhong, A., Cao, Y., 2022. Greenium in the Chinese corporate bond market. Emerg. Mark. Rev. 53, 100946.
Hyun, S., Park, D., Tian, S., 2021. Pricing of green labeling: a comparison of labeled and unlabeled green bonds. Finance Res. Lett. 41, 101816.
Imbens, G.W., Wooldridge, J.M., 2009. Recent developments in the econometrics of program evaluation. J. Econ. Lit. 47 (1), 5–86.
Immel, M., Hachenberg, B., Kiesel, F., Schiereck, D., 2021. Green bonds: shades of green and brown. J. Asset Manag. 22 (2), 96–109.
International Capital Markets Association, 2021. Green Bond Principles: Voluntary Process Guidelines for Issuing Green Bonds.
Kapraun, J., Latino, C., Scheins, C., Schlag, C., 2021. (In)-credibly green: which bonds trade at a green bond premium? In: Proceedings of Paris December 2019 Finance Meeting EUROFIDAI-ESSEC.
Karpf, A., Mandel, A., 2018. The changing value of the 'green' label on the U.S. municipal bond market. Nat. Clim. Change 8 (2), 161–165.
Krishnamurthy, A., 2002. The bond/old-bond spread. J. Financ. Econ. 66 (2–3), 463–506.
Larcker, D.F., Watts, E.M., 2020. Where's the greenium? J. Account. Econ. 69.
Lau, P., Sze, A., Wan, W., Wong, A., 2022. The economics of the greenium: how much is the world willing to pay to save the Earth? Environ. Resour. Econ. 81 (2), 379–408.
Li, Q., Zhang, K., Wang, L., 2022. Where's the green bond premium? Evidence from China. Finance Res. Lett. 48, 102950.
Lin, J.-D., 2023. Explaining the quality of green bonds in China. J. Clean. Prod. 406, 136893.
Ma, Y., 2019. Nonfinancial firms as cross-market arbitrageurs. J. Finance 74 (6), 3041–3087.
Marsh, A., 2020. Green Bond Crisis Premium Theory Debunked by Covid, Manager Says. Bloomberg.
Miranda-Agrippino, S., Nenova, T., Rey, H., 2020. Global Footprints of Monetary Policies. CFM, Centre for Macroeconomics, London.
Morningstar, Inc., 2021. Morningstar Direct.
Mota, L., 2020. The corporate supply of (quasi) safe assets. Working Paper (SSRN 3732444).
Nanayakkara, M., Colombage, S., 2019. Do investors in green bond market pay a premium? Global evidence. Appl. Econ. 51 (40), 4425–4437.
Nikolova, S., Wang, L., Wu, J.J., 2020. Institutional allocations in the primary market for corporate bonds. J. Financ. Econ. 137 (2), 470–490.
Oikonomou, I., Brooks, C., Pavelin, S., 2014. The effects of corporate social performance on the cost of corporate debt and credit ratings. Financ. Rev. 49 (1), 49–75.
Pandolfi, L., Williams, T., 2019. Capital flows and sovereign debt markets: evidence from index rebalancings. J. Financ. Econ. 132 (2), 384–403.
Partnership for Carbon Accounting Financials, 2021. Draft: New methods for public consultation: for financial institutions measuring and reporting scope 3 category 15 emissions.
Partridge, C., Medda, F.R., 2020. Green bond pricing: the search for greenium. J. Altern. Invest. 23.
Pietsch, A., Salakhova, D., 2022. Pricing of green bonds: drivers and dynamics of the greenium.
Quinson, T., 2021a. How Wall Street Is Gaming ESG Scores. Bloomberg.
Quinson, T., 2021b. Regulators Intensify ESG Scrutiny as Greenwashing Explodes. Bloomberg.
Raddatz, C., Schmukler, S.L., Williams, T., 2017. International asset allocations and capital flows: the benchmark effect. J. Int. Econ. 108, 413–430.
Refinitiv Eikon Premium Kiosks, 2021. Refinitiv.
Refinitiv Workspace, 2021. Refinitiv.
Ritchie, G., Ward, J., Kishan, S., Gledhill, A., 2021. Bond Investor Revolt Brews over Bogus Green Debt Flooding Market. Bloomberg.
Rosenbaum, P.R., 2020. Modern algorithms for matching in observational studies. Annu. Rev. Stat. Appl. 7, 143–176.
Siani, K.Y., 2022. Raising bond capital in segmented markets. Working Paper.
Stellner, C., Klein, C., Zwergel, B., 2015. Corporate social responsibility and Eurozone corporate bonds: the moderating role of country sustainability. J. Bank. Finance 59, 538–549.
Stubbington, T., 2021. Squeeze on 'Greenium' as ESG Bond Investors Demand More Value. Financial Times.
Stubbington, T., Nauman, B., 2020. Investors probe ESG credentials of bond sellers on 'greenwashing' fears. Financ. Times.
Tang, D.Y., Zhang, Y., 2020. Do shareholders benefit from green bonds? J. Corp. Finance 61.
Tang, Y., Wang, B., Pan, N., Li, Z., 2023. The impact of environmental information disclosure on the cost of green bond: evidence from China. Energy Econ. 126, 107008.
Temple-West, P., Palma, S., 2022. SEC prepares to crack down on misleading ESG investment claims. Financ. Times.
The Economist, 2021. Sustainable finance is rife with greenwash. Time for more disclosure.
Vayanos, D., Weill, P.-O., 2008. A search-based theory of the on-the-run phenomenon. J. Finance 63 (3), 1361–1398.
Wang, J., Chen, X., Li, X., Yu, J., Zhong, R., 2020. The market reaction to green bond issuance: evidence from China. Pac.-Basin Finance J. 60, 101294.
Wang, L., Wu, J., 2023. Deciphering greenium: the role of investor demand. University of Nebraska-Lincoln Working Paper.
Ye, S., 2019. How do ETFs affect the liquidity of the underlying corporate bonds. Chinese University of Hong Kong Working Paper.
Zerbib, O.D., 2019. The effect of pro-environmental preferences on bond prices: evidence from green bonds. J. Bank. Finance 98, 39–60.